

# ISC Class XII Mathematics

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## Chapter 11 Three Dimensional Geometry

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### 64 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

#### Sections Covered

- I. Direction Cosines & Direction Ratios (Types 1–16)
- II. Equation of a Line in Space (Types 17–29)
- III. Angle Between Lines & Shortest Distance (Types 30–38)
- IV. The Plane: Forms & Equations (Types 39–56)
- V. Angle Between Planes & Distance Formulas (Types 57–64)

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| Type                                                             | Method / Pattern                                                 | ★   | Source(s)          |
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| Type                                                            | Method / Pattern                                                  | ★   | Source(s)            |
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| Type | Method / Pattern                                          | ★   | Source(s)            |
|------|-----------------------------------------------------------|-----|----------------------|
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| 63   | Distance between two parallel planes; mid-parallel plane  | –   | Ex 11.8 Q5, Q16, Q17 |
| 64   | Distance of a point from a plane, measured along a line   | –   | Ex 11.8 Q20          |

## Formula Sheet

### Direction Cosines & Ratios

#### Basic identity

$$l^2 + m^2 + n^2 = 1$$

$l, m, n$  = direction cosines of a line

#### DRs to DCs

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$a, b, c$  = direction ratios (any multiple of DCs)

#### DCs of line joining two points

$$l = \frac{x_2 - x_1}{PQ}, \quad m = \frac{y_2 - y_1}{PQ}, \quad n = \frac{z_2 - z_1}{PQ}$$

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

#### Angle between two lines

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2 = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Use DCs directly, or DRs with the fraction form

#### Condition for perpendicularity

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Equivalently  $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

#### Condition for parallelism

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

DRs of one line are proportional to the other

## The Line in Space

### Vector equation (point + direction)

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

$\vec{a}$  = position vector of a known point,  $\vec{b}$  = direction vector

### Cartesian equation (point + direction)

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

$(x_1, y_1, z_1)$  on the line;  $a, b, c$  = DRs

### Vector equation (through two points)

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

$\vec{a}, \vec{b}$  = position vectors of the two points

### Cartesian equation (through two points)

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Joining  $(x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$

### Point at parameter distance $r$

$$(x_1 + lr, y_1 + mr, z_1 + nr)$$

Use DCs (not raw DRs) so that  $r$  is an actual distance

## Shortest Distance Between Lines

### S.D. between skew lines (vector form)

$$\text{S.D.} = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

Lines  $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$  and  $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$

### S.D. between parallel lines

$$\text{S.D.} = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$$

Both lines share direction vector  $\vec{b}$

**Lines intersect iff**

$$[\vec{a}_2 - \vec{a}_1 \quad \vec{b}_1 \quad \vec{b}_2] = 0$$

Scalar triple product vanishes – S.D. = 0

**The Plane****Vector equation (normal form)**

$$\vec{r} \cdot \hat{n} = d$$

$\hat{n}$  = unit normal,  $d$  = distance from origin

**Cartesian equation (point + normal)**

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

Plane through  $(x_1, y_1, z_1)$  with normal DRs  $\langle a, b, c \rangle$

**Intercept form**

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$a, b, c$  = intercepts on the  $x, y, z$  axes

**Plane through three points**

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$  are the three points

**Plane through intersection of two planes**

$$P_1 + \lambda P_2 = 0$$

$P_1 = 0, P_2 = 0$  are the two given planes;  $\lambda$  found from the extra condition

**Plane parallel to a given plane**

$$ax + by + cz = \lambda$$

Same normal  $\langle a, b, c \rangle$ ;  $\lambda$  fixed by the point it must pass through

**Angles & Distances**

**Angle between two planes**

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Angle between the normals  $\vec{n}_1, \vec{n}_2$

**Perpendicularity of two planes**

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\vec{n}_1 \cdot \vec{n}_2 = 0$$

**Angle between a line and a plane**

$$\sin \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Line DRs  $\langle a_1, b_1, c_1 \rangle$ , plane normal  $\langle a_2, b_2, c_2 \rangle$  – note sin, not cos

**Distance of a point from a plane**

$$d = \left| \frac{ax_1 + by_1 + cz_1 - k}{\sqrt{a^2 + b^2 + c^2}} \right|$$

Point  $(x_1, y_1, z_1)$ ; plane  $ax + by + cz = k$

**Distance between parallel planes**

$$d = \left| \frac{k_1 - k_2}{\sqrt{a^2 + b^2 + c^2}} \right|$$

Planes  $ax + by + cz = k_1$  and  $ax + by + cz = k_2$

**Scalar triple product**

$$[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Zero  $\iff$  the three vectors are coplanar

## I. Direction Cosines & Direction Ratios

### Type 1

Finding the distance of a point from a coordinate axis, or from a coordinate plane. The foot of the perpendicular on an axis keeps only that axis's coordinate; the foot of the perpendicular on a coordinate plane drops the coordinate not in that plane's name.

*Source: Exercise 11.1, Q1(i),(iii)*

**Given:** (a) Point  $P(2, -3, 4)$ , distance from the  $x$ -axis.      (b) Point  $P(5, -6, 3)$ , distance from the XOY plane.

**Solution:**

#### Case: (a) Distance from the $x$ -axis

The foot of perpendicular from  $P(2, -3, 4)$  on the  $x$ -axis keeps the  $x$ -coordinate and zeroes the rest, so  $M = (2, 0, 0)$ .

Required distance  $= |PM| = \sqrt{(2-2)^2 + (-3-0)^2 + (4-0)^2} = \sqrt{9+16} = 5$  units, since PM is just the ordinary distance formula between P and M.

#### Case: (b) Distance from the XOY (i.e. $z = 0$ ) plane

Every point's perpendicular distance from the plane  $z = 0$  is simply the magnitude of its  $z$ -coordinate, because that plane is defined by  $z = 0$ .

Required distance  $= |3| = 3$  units.

#### Final Answer

(a) 5 units      (b) 3 units

**Common Mistake:** Students often drop the wrong coordinates when finding the foot of perpendicular on an axis – remember only the coordinate matching the axis name survives, the other two become 0.

**Quick Recall:** Distance from an axis = keep that axis's coordinate, zero the rest, then use distance formula; distance from a coordinate plane = just the absolute value of the missing coordinate.

**Type 2**

Finding direction cosines of a line when its angles with the coordinate axes are given (or partly given). Uses  $l = \cos \alpha$ ,  $m = \cos \beta$ ,  $n = \cos \gamma$  together with the identity  $l^2 + m^2 + n^2 = 1$  to find any missing angle.

Source: Exercise 11.1, Q2(i), Q5(i)

**Given:** (a) A line makes angles  $90^\circ, 60^\circ, 30^\circ$  with the positive  $x, y, z$  axes. (b) A line makes angles  $90^\circ, 60^\circ$  with  $x, y$  axes and  $\theta$  (acute) with the  $z$ -axis; find  $\theta$ .

**Solution:**

**Case: (a) Direct application**

Since  $l = \cos \alpha$ ,  $m = \cos \beta$ ,  $n = \cos \gamma$  by definition of direction cosines,  
 $l = \cos 90^\circ = 0$ ,  $m = \cos 60^\circ = \frac{1}{2}$ ,  $n = \cos 30^\circ = \frac{\sqrt{3}}{2}$ .

**Case: (b) Using the identity to find a missing angle**

Here  $l = \cos 90^\circ = 0$ ,  $m = \cos 60^\circ = \frac{1}{2}$ . Since every line satisfies  $l^2 + m^2 + n^2 = 1$ , we substitute the known values to solve for  $n$ .

$$0 + \frac{1}{4} + n^2 = 1 \Rightarrow n^2 = \frac{3}{4} \Rightarrow n = \pm \frac{\sqrt{3}}{2}.$$

As  $\theta$  is acute,  $\cos \theta > 0$ , so we take  $n = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$ .

**Final Answer**

(a)  $\left\langle 0, \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$  (b)  $\theta = 30^\circ$

**Common Mistake:** When solving  $n^2 = \frac{3}{4}$ , students forget the  $\pm$  and then forget to actually apply the "acute angle" condition to choose the correct sign.

**Quick Recall:** DCs are just the cosines of the angles with each axis; if two angles are known, the identity  $l^2 + m^2 + n^2 = 1$  pins down the third.

**Type 3**

Checking whether a given triple of numbers (or a given triple of angles) can actually be the direction cosines of some real line. The test is always the identity  $l^2 + m^2 + n^2 = 1$ .

Source: Exercise 11.1, Q3(i), (iii)

**Given:** (a) Can  $\left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$  be DCs of a line? (b) Prove  $1, 1, 1$  cannot be DCs of a line.

**Solution:**

**Case: (a)**

We test the identity:  $l^2 + m^2 + n^2 = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} \neq 1$ , so the identity fails.

**Case: (b)**

Similarly,  $1^2 + 1^2 + 1^2 = 3 \neq 1$ . Since every genuine set of direction cosines must satisfy  $l^2 + m^2 + n^2 = 1$  and  $3 \neq 1$ , no line can have 1, 1, 1 as its direction cosines.

**Final Answer**

(a) No (b) Proved – 1, 1, 1 fail the identity, hence cannot be DCs

**Common Mistake:** Confusing direction *ratios* (which need no such restriction) with direction *cosines* (which must satisfy  $l^2 + m^2 + n^2 = 1$ ) – any three numbers can be DRs, but not DCs.

**Quick Recall:** To check if numbers can be DCs, square and add them – if the sum isn't exactly 1, they cannot be direction cosines of any line.

**Type 4**

DCs of a line are given as equal multiples of a single unknown constant; find the constant using  $l^2 + m^2 + n^2 = 1$ .

Source: Exercise 11.1, Q4

**Given:** Direction cosines of a line are  $\langle \frac{1}{c}, \frac{1}{c}, \frac{1}{c} \rangle$ . Find  $c$ .

**Solution:**

Since these are stated to be actual direction cosines, they must satisfy the identity  $l^2 + m^2 + n^2 = 1$ .  
 $\frac{1}{c^2} + \frac{1}{c^2} + \frac{1}{c^2} = 1 \Rightarrow \frac{3}{c^2} = 1 \Rightarrow c^2 = 3 \Rightarrow c = \pm\sqrt{3}$ , taking the square root on both sides and keeping both signs since  $c$  itself was not restricted to be positive.

**Final Answer**

$$c = \pm\sqrt{3}$$

**Common Mistake:** Dropping the  $\pm$  when taking the square root of  $c^2 = 3$ .

**Quick Recall:** Whenever DCs are given in terms of an unknown, plug straight into  $l^2 + m^2 + n^2 = 1$  and solve.

**Type 5**

Converting a given set of direction ratios into direction cosines by dividing each ratio by the magnitude  $\sqrt{a^2 + b^2 + c^2}$ .

**Formula Used:**  $l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}, m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$

Source: Exercise 11.1, Q6(i)

**Given:** Direction ratios of a line are 2, -1, -2.

**Solution:**

First find the magnitude, since DCs are DRs divided by this magnitude:  $\sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$ .

Dividing each ratio by 3 gives  $l = \frac{2}{3}, m = -\frac{1}{3}, n = -\frac{2}{3}$ .

**Final Answer**

$$\left\langle \frac{2}{3}, -\frac{1}{3}, -\frac{2}{3} \right\rangle$$

**Common Mistake:** Forgetting to square-root the sum of squares – i.e. dividing by  $a^2 + b^2 + c^2$  instead of  $\sqrt{a^2 + b^2 + c^2}$ .

**Quick Recall:** DCs = DRs divided by the square root of the sum of their squares.

**Type 6**

Finding the direction ratios and direction cosines of the line joining two given points, using the coordinate-difference method.

**Formula Used:** DRs =  $\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$

Source: Exercise 11.1, Q7(ii)

**Given:** Points P(2, 3, 5) and Q(-1, 2, 4).

**Solution:**

The direction ratios of the line PQ are obtained by subtracting corresponding coordinates (Q - P), since the line's direction is the displacement from one point to the other:

$$\langle -1 - 2, 2 - 3, 4 - 5 \rangle = \langle -3, -1, -1 \rangle.$$

Now find the magnitude to convert to DCs:  $PQ = \sqrt{(-3)^2 + (-1)^2 + (-1)^2} = \sqrt{9 + 1 + 1} = \sqrt{11}$ .

Dividing through,  $l = \frac{-3}{\sqrt{11}}, m = \frac{-1}{\sqrt{11}}, n = \frac{-1}{\sqrt{11}}$ .

**Final Answer**

DRs  $\langle -3, -1, -1 \rangle$ ; DCs  $\left\langle \frac{-3}{\sqrt{11}}, \frac{-1}{\sqrt{11}}, \frac{-1}{\sqrt{11}} \right\rangle$

**Common Mistake:** Subtracting in the wrong order ( $P - Q$  instead of  $Q - P$ ) – this only flips every sign, but students then get inconsistent signs when combining with other given data.

**Quick Recall:** Direction ratios of a line joining two points = (second point) minus (first point), coordinate by coordinate.

**Type 7**

Showing that two (or three) lines are mutually perpendicular using direction ratios: the lines are perpendicular exactly when the dot product of their DRs is zero.

★ **Board Favorite** – A very standard "show that" pattern – appears almost every year in some form, either for a single pair or for three mutually perpendicular lines together.

**Formula Used:**  $a_1a_2 + b_1b_2 + c_1c_2 = 0 \iff$  lines are perpendicular

Source: Exercise 11.1, Q13

**Given:** Show the lines with DCs  $\left\langle \frac{12}{13}, -\frac{3}{13}, -\frac{4}{13} \right\rangle$ ,  $\left\langle \frac{4}{13}, \frac{12}{13}, \frac{3}{13} \right\rangle$ ,  $\left\langle \frac{3}{13}, -\frac{4}{13}, \frac{12}{13} \right\rangle$  are mutually perpendicular.

**Solution:**

Since there are three lines, we must check all  $\binom{3}{2} = 3$  pairs, and each pair is perpendicular exactly when its dot product vanishes.

$$\text{Pair 1 \& 2: } \frac{12 \cdot 4}{169} + \frac{(-3)(12)}{169} + \frac{(-4)(3)}{169} = \frac{48 - 36 - 12}{169} = \frac{0}{169} = 0.$$

$$\text{Pair 2 \& 3: } \frac{4 \cdot 3}{169} + \frac{12 \cdot (-4)}{169} + \frac{3 \cdot 12}{169} = \frac{12 - 48 + 36}{169} = 0.$$

$$\text{Pair 1 \& 3: } \frac{12 \cdot 3}{169} + \frac{(-3)(-4)}{169} + \frac{(-4)(12)}{169} = \frac{36 + 12 - 48}{169} = 0.$$

All three dot products vanish, so every pair is perpendicular.

**Final Answer**

All three pairs give dot product 0 – hence the lines are mutually perpendicular.

**Common Mistake:** Checking only one pair and assuming the other two follow automatically – with three lines you must verify all three pairs separately.

**Quick Recall:** Dot product of DRs = 0 means perpendicular; for three lines "mutually perpendicular" needs all three pairwise checks.

### Type 8

Finding an unknown value so that two lines (each given through a pair of points) become perpendicular, or so that they become parallel.

Source: Exercise 11.1, Q9(i),(ii)

**Given:** (a) For what  $p$  is the line through  $(4, 1, 2)$  and  $(5, p, 0)$  perpendicular to the line through  $(2, 1, 1)$  and  $(3, 3, -1)$ ? (b) Find  $p, q$  so the line joining  $(7, p, 2)$  and  $(q, -2, 5)$  is parallel to the line joining  $(2, -3, 5)$  and  $(-6, -15, 11)$ .

**Solution:**

#### Case: (a) Perpendicularity

DRs of the first line:  $\langle 5 - 4, p - 1, 0 - 2 \rangle = \langle 1, p - 1, -2 \rangle$ .

DRs of the second line:  $\langle 3 - 2, 3 - 1, -1 - 1 \rangle = \langle 1, 2, -2 \rangle$ .

Perpendicular lines need the dot product of their DRs to be zero:  $1(1) + (p - 1)(2) + (-2)(-2) = 0 \Rightarrow 1 + 2p - 2 + 4 = 0 \Rightarrow 2p + 3 = 0 \Rightarrow p = -\frac{3}{2}$ .

#### Case: (b) Parallelism

DRs of the first line:  $\langle q - 7, -2 - p, 3 \rangle$ . DRs of the second:  $\langle -6 - 2, -15 + 3, 11 - 5 \rangle = \langle -8, -12, 6 \rangle$ , i.e. simplified  $\langle -4, -6, 3 \rangle$ .

For parallel lines, the DRs must be proportional, so we match the  $z$ -ratio first since it is already numeric:  $\frac{3}{3} = 1$ , meaning the whole first triple must equal the second triple exactly (since the second has  $z = 3$ , same scale).

So  $q - 7 = -4 \Rightarrow q = 3$ , and  $-2 - p = -6 \Rightarrow p = 4$ .

### Final Answer

(a)  $p = -\frac{3}{2}$  (b)  $p = 4, q = 3$

**Common Mistake:** In the parallel case, students often set up the proportion but pick the wrong two ratios to equate first, leading to messy fractions instead of the clean scale factor.

**Quick Recall:** Perpendicular  $\Rightarrow$  dot product of DRs is 0; parallel  $\Rightarrow$  DRs are proportional – match the ratios coordinate-wise.

### Type 9

Showing that three given points are collinear by computing the direction ratios of two segments formed from them and checking that these DRs are proportional.

Source: Exercise 11.1, Q10

**Given:** Show that  $(2, 3, 4)$ ,  $(-1, -2, 1)$ ,  $(5, 8, 7)$  are collinear.

**Solution:**

Let  $A(2, 3, 4)$ ,  $B(-1, -2, 1)$ ,  $C(5, 8, 7)$ . We find DRs of AB and BC; if they are proportional, the three points lie on one straight line.

DRs of AB =  $\langle -1 - 2, -2 - 3, 1 - 4 \rangle = \langle -3, -5, -3 \rangle$ .

DRs of BC =  $\langle 5 - (-1), 8 - (-2), 7 - 1 \rangle = \langle 6, 10, 6 \rangle$ .

Notice  $\langle 6, 10, 6 \rangle = -2 \times \langle -3, -5, -3 \rangle$ , so the DRs of BC are a scalar multiple of the DRs of AB, meaning AB and BC are parallel. Since they also share the common point B, the three points must be collinear.

**Final Answer**

DRs of AB and BC are proportional  $(-2 : 1)$  and share point B  $\Rightarrow$  A, B, C are collinear.

**Common Mistake:** Showing the DRs are proportional but forgetting to note the shared point B – proportional DRs alone only prove parallel lines, the common point is what upgrades this to collinearity.

**Quick Recall:** Collinear  $\Rightarrow$  DRs of any two segments among the three points are proportional AND they share a point.

### Type 10

Finding the angle between two lines when their direction cosines, direction ratios, or direction ratios in surd/mixed form are given directly (not via points). Uses the standard cosine formula.

★ **Board Favorite** – Appears in nearly every board paper in one of its three input forms (DCs / DRs / surd ratios) – one of the highest-yield formulas in the whole chapter.

**Formula Used:** 
$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Source: Exercise 11.1, Q12(ii)

**Given:** Find the angle between lines with direction ratios  $\langle 1, 2, 1 \rangle$  and  $\langle 2, -3, 4 \rangle$ .

**Solution:**

Substituting directly into the cosine formula: numerator =  $1(2) + 2(-3) + 1(4) = 2 - 6 + 4 = 0$ .

Since the numerator (the dot product) is already 0, we don't even need the denominator –  $\cos \theta = 0$  immediately gives  $\theta = 90^\circ$ .

### Final Answer

$$\theta = 90^\circ$$

**Common Mistake:** Continuing to compute the magnitudes and simplify a fraction even when the numerator is already 0 – a wasted step; always check the numerator first.

### Alternate Board-Style Example

Source: Exercise 11.1, Q12(i)

**Given:** Lines with DCs  $\left\langle \frac{\sqrt{3}}{2}, \frac{1}{4}, \frac{\sqrt{3}}{4} \right\rangle$  and  $\left\langle -\frac{\sqrt{3}}{2}, \frac{1}{4}, \frac{\sqrt{3}}{4} \right\rangle$ .

### Solution:

Since these are already unit DCs,  $\cos \theta$  is simply their dot product (no denominator needed):

$$\cos \theta = \left( \frac{\sqrt{3}}{2} \right) \left( -\frac{\sqrt{3}}{2} \right) + \left( \frac{1}{4} \right) \left( \frac{1}{4} \right) + \left( \frac{\sqrt{3}}{4} \right) \left( \frac{\sqrt{3}}{4} \right) = -\frac{3}{4} + \frac{1}{16} + \frac{3}{16} = -\frac{12}{16} + \frac{4}{16} = -\frac{1}{2}.$$

$\theta = \cos^{-1} \left( -\frac{1}{2} \right) = 120^\circ$ ; if the acute angle between the lines is wanted, we instead report  $60^\circ$ .

### Final Answer

$\theta = 120^\circ$  (or the acute angle  $60^\circ$ , depending on what is asked)

**Quick Recall:** Angle between lines: dot the DRs, divide by the product of their magnitudes; if DCs are already given, skip the denominator entirely.

### Type 11

Finding the angle between two lines when each line is specified only as a pair of points – first convert each pair of points to DRs, then apply the standard angle formula.

Source: Exercise 11.1, Q15

**Given:** A(1, 2, 3), B(4, 5, 7), C(-4, 3, -6), D(2, 9, 2). Find the angle between lines AB and CD.

### Solution:

DRs of AB =  $\langle 4 - 1, 5 - 2, 7 - 3 \rangle = \langle 3, 3, 4 \rangle$ .

DRs of CD =  $\langle 2 - (-4), 9 - 3, 2 - (-6) \rangle = \langle 6, 6, 8 \rangle$ .

Notice CD's DRs =  $2 \times$  AB's DRs, i.e. the two direction vectors are parallel (scalar multiples of each other), so lines AB and CD are parallel and the angle between them is  $0^\circ$ .

**Final Answer**

$\theta = 0^\circ$  (the lines are parallel)

**Common Mistake:** Blindly plugging into the cosine formula without first noticing the DRs are already proportional – computing this way still gives  $\cos \theta = 1$ , but recognising proportionality first is faster and less error-prone.

**Quick Recall:** Two points  $\rightarrow$  subtract to get DRs, then use the usual angle formula; always scan for proportionality first, since it may reveal the lines are parallel instantly.

**Type 12**

Showing that four given points form a parallelogram by proving opposite sides have equal and parallel direction ratios.

Source: Exercise 11.1, Q16

**Given:** A(4, 7, 8), B(2, 3, 4), C(-1, -2, 1), D(1, 2, 5), taken in order.

**Solution:**

For ABCD to be a parallelogram, side AB must be parallel and equal to side DC (not CD, since we need same direction going "forward" around the shape).

DRs of AB =  $\langle 2 - 4, 3 - 7, 4 - 8 \rangle = \langle -2, -4, -4 \rangle$ .

DRs of DC =  $\langle -1 - 1, -2 - 2, 1 - 5 \rangle = \langle -2, -4, -4 \rangle$ .

Since AB and DC have identical direction ratios, AB is parallel to DC and (because the ratios are exactly equal, not just proportional) also equal in length. A quadrilateral with one pair of sides both parallel and equal is a parallelogram.

**Final Answer**

AB  $\parallel$  DC and  $|AB| = |DC| \Rightarrow$  ABCD is a parallelogram.

**Common Mistake:** Comparing AB with CD instead of DC – this reverses the direction and makes the DRs negatives of each other, which still shows parallel but can confuse the "equal length, correctly oriented" argument.

**Quick Recall:** Parallelogram proof: show one pair of opposite sides (matched in traversal direction) has identical direction ratios.

**Type 13**

Finding the direction cosines of a line that is perpendicular to two other given lines, using the cross product of their direction ratios.

**Formula Used:**  $\vec{n} = \vec{b}_1 \times \vec{b}_2$

Source: Exercise 11.1, Q17

**Given:** Find the DCs of a line perpendicular to the two lines joining A(2, 3, -4) to B(-3, 3, -2) and C(-1, 4, 2) to D(3, 5, 1).

**Solution:**

DRs of AB =  $\langle -5, 0, 2 \rangle$ ; DRs of CD =  $\langle 4, 1, -1 \rangle$ . A line perpendicular to both must be along their cross product, because the cross product of two vectors is perpendicular to each of them.

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 0 & 2 \\ 4 & 1 & -1 \end{vmatrix} = \hat{i}(0 \cdot (-1) - 2 \cdot 1) - \hat{j}((-5)(-1) - 2 \cdot 4) + \hat{k}((-5)(1) - 0 \cdot 4), \text{ expanding}$$

along the top row.

$$= \hat{i}(-2) - \hat{j}(5 - 8) + \hat{k}(-5) = -2\hat{i} + 3\hat{j} - 5\hat{k}, \text{ i.e. DRs } \langle -2, 3, -5 \rangle.$$

Magnitude =  $\sqrt{4 + 9 + 25} = \sqrt{38}$ , so dividing through gives the DCs.

**Final Answer**

$$\left\langle \frac{-2}{\sqrt{38}}, \frac{3}{\sqrt{38}}, \frac{-5}{\sqrt{38}} \right\rangle$$

**Common Mistake:** Sign errors while expanding the  $3 \times 3$  determinant for the cross product – always double-check the middle term's sign, it's the one most often dropped.

**Quick Recall:** A line perpendicular to two other lines has direction along the cross product of their DRs.

### Type 14

Finding the angle between two lines whose direction cosines are only known to satisfy two linear/homogeneous relations – eliminate one variable between the relations, substitute into  $l^2 + m^2 + n^2 = 1$ , and solve the resulting quadratic to get the actual DCs (or directly the angle) of both lines.

★ **Board Favorite** – A signature ISC-exemplar technique – shows up whenever DCs are given implicitly through equations rather than numbers, testing algebraic elimination skill on top of the 3D geometry idea.

Source: Exercise 11.1, Q19(ii)

**Given:** Find the angle between the lines whose DCs satisfy  $3l + m + 5n = 0$  and  $6mn - 2nl + 5lm = 0$ .

**Solution:**

From the linear relation,  $m = -3l - 5n$ . We substitute this into the second (homogeneous quadratic) relation to eliminate  $m$ , since that reduces everything to a relation purely between  $l$  and  $n$ .

$6n(-3l - 5n) - 2nl + 5l(-3l - 5n) = 0 \Rightarrow -18nl - 30n^2 - 2nl - 15l^2 - 25nl = 0$   
 $\Rightarrow -15l^2 - 45nl - 30n^2 = 0 \Rightarrow l^2 + 3nl + 2n^2 = 0$  (dividing through by  $-15$ ), which factorises as  $(l + n)(l + 2n) = 0$ .

**Case:**  $l = -n$

Then from  $m = -3l - 5n = -3(-n) - 5n = 3n - 5n = -2n$ . So DRs are  $\langle -n, -2n, n \rangle$ , i.e.  $\langle -1, -2, 1 \rangle$  after cancelling  $n$ .

**Case:**  $l = -2n$

Then  $m = -3(-2n) - 5n = 6n - 5n = n$ . So DRs are  $\langle -2n, n, n \rangle$ , i.e.  $\langle -2, 1, 1 \rangle$ .

Now the angle between the two actual lines (DRs  $\langle -1, -2, 1 \rangle$  and  $\langle -2, 1, 1 \rangle$ ) is found by the usual formula:

$$\cos \theta = \frac{(-1)(-2) + (-2)(1) + (1)(1)}{\sqrt{1+4+1}\sqrt{4+1+1}} = \frac{2-2+1}{\sqrt{6}\sqrt{6}} = \frac{1}{6}.$$

### Final Answer

$$\theta = \cos^{-1} \left( \frac{1}{6} \right)$$

**Common Mistake:** After factorising into  $(l + n)(l + 2n) = 0$ , forgetting that each root gives a *separate* line's DRs – both cases must be carried through to get two full direction-ratio triples before the angle formula can be applied.

**Quick Recall:** Two DC relations  $\rightarrow$  eliminate one variable, factor the resulting quadratic in the ratio of the remaining two, and each factor hands you the DRs of one of the two lines.

### Type 15

Finding the foot of the perpendicular from a given point (possibly the origin) onto the line joining two other points, and, as a follow-up, the image (reflection) of that point in the line.

★ **Board Favorite** – This exact worked pattern opens the chapter in ML Aggarwal and is one of the most repeated long-answer questions in ISC boards – master this one carefully.

**Formula Used:**  $N = \left( \frac{2\lambda + 1}{\lambda + 1}, \dots \right)$  from the section-formula point on BC, then  $\vec{AN} \cdot \vec{BC} = 0$

Source: Exercise 11.1, Q21

**Given:** Find the foot of perpendicular from A(1, 2, 1) on the line joining B(1, 4, 6) and C(5, 4, 4).

**Solution:**

Let N divide BC in ratio  $\lambda : 1$ . By the section formula,  $N = \left( \frac{5\lambda + 1}{\lambda + 1}, \frac{4\lambda + 4}{\lambda + 1}, \frac{4\lambda + 6}{\lambda + 1} \right)$ , since this is how any point dividing BC in ratio  $\lambda : 1$  is written.

DRs of AN =  $\left\langle \frac{5\lambda + 1}{\lambda + 1} - 1, \frac{4\lambda + 4}{\lambda + 1} - 2, \frac{4\lambda + 6}{\lambda + 1} - 1 \right\rangle = \left\langle \frac{4\lambda}{\lambda + 1}, \frac{2\lambda + 2}{\lambda + 1}, \frac{3\lambda + 5}{\lambda + 1} \right\rangle$ , simplifying each fraction.

DRs of BC =  $\langle 5 - 1, 4 - 4, 4 - 6 \rangle = \langle 4, 0, -2 \rangle$ , i.e.  $\langle 2, 0, -1 \rangle$ .

Since AN must be perpendicular to BC (N is the foot of perpendicular), their dot product is zero:

$$4\lambda(2) + (2\lambda + 2)(0) + (3\lambda + 5)(-1) = 0 \Rightarrow 8\lambda - 3\lambda - 5 = 0 \Rightarrow 5\lambda = 5 \Rightarrow \lambda = 1.$$

Substituting  $\lambda = 1$  back into N:  $N = \left( \frac{6}{2}, \frac{8}{2}, \frac{10}{2} \right) = (3, 4, 5)$ .

**Final Answer**

Foot of perpendicular N = (3, 4, 5)

**Common Mistake:** Forgetting to multiply through by  $(\lambda + 1)$  before taking the dot product – carrying the fractions through the perpendicularity equation invites arithmetic slips.

**Alternate Board-Style Example (image of a point)**

Source: Exercise 11.1, Q20

**Given:** Find the point in which the join of A(-9, 4, 5) and B(11, 0, -1) is met by the perpendicular from the origin O.

**Solution:**

Let N divide AB in ratio  $\lambda : 1$ , so  $N = \left( \frac{11\lambda - 9}{\lambda + 1}, \frac{4}{\lambda + 1}, \frac{-\lambda + 5}{\lambda + 1} \right)$ .

DRs of ON = the coordinates of N themselves (since O is the origin); DRs of AB =  $\langle 20, -4, -6 \rangle$ , i.e.  $\langle 10, -2, -3 \rangle$ .

ON  $\perp$  AB gives  $10(11\lambda - 9) - 2(4) - 3(-\lambda + 5) = 0$  (after clearing the common denominator, which doesn't affect the zero condition).

$$110\lambda - 90 - 8 + 3\lambda - 15 = 0 \Rightarrow 113\lambda = 113 \Rightarrow \lambda = 1.$$

So N =  $\left( \frac{2}{2}, \frac{4}{2}, \frac{4}{2} \right) = (1, 2, 2)$ .

**Final Answer**

N = (1, 2, 2)

**Quick Recall:** Put the foot of perpendicular as a section-formula point on the given line, force perpendicularity with the line's DRs to solve for the ratio, then substitute back; for an image, the foot is the midpoint of the original point and its reflection.

**Type 16**

A pure proof-type sum: showing that the angle between any two of the four space-diagonals of a rectangular box (with edges  $a, b, c$  along the axes) is given by a stated general cosine formula, using the DCs of each diagonal.

Source: Exercise 11.1, Q22

**Given:** A rectangular box has edges  $a, b, c$  along the  $x, y, z$  axes. Prove the angle between any two of its four diagonals is  $\cos^{-1} \left( \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$  where the signs cannot all be the same.

**Solution:**

Place one vertex at the origin  $O(0, 0, 0)$ , with the box extending to  $(a, 0, 0), (0, b, 0), (0, 0, c), (a, b, 0), (a, 0, c), (0, b, c), (a, b, c)$ , so all eight vertices are labelled by which of  $a, b, c$  they include.

The diagonal from  $O$  to the opposite vertex  $(a, b, c)$  has DRs  $\langle a, b, c \rangle$ .

A diagonal starting from a different vertex, e.g.  $(a, 0, 0)$  to  $(0, b, c)$ , has DRs  $\langle 0 - a, b - 0, c - 0 \rangle = \langle -a, b, c \rangle$  – one sign has flipped compared to the first diagonal, because this diagonal travels in the opposite sense along the  $x$ -edge.

By the angle formula between two lines using these DRs,  $\cos \theta = \frac{(a)(-a) + (b)(b) + (c)(c)}{\sqrt{a^2 + b^2 + c^2} \sqrt{a^2 + b^2 + c^2}} = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}$ .

Each of the four diagonals corresponds to one particular sign pattern on  $a^2, b^2, c^2$  (never all the same sign, since that would make two diagonals coincide as the same line), which is exactly the general form  $\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}$  to be proved.

**Final Answer**

Proved: angle between diagonals  $= \cos^{-1} \left( \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$ , signs not all identical.

**Common Mistake:** Forgetting that the denominator is always  $a^2 + b^2 + c^2$  with all  $+$  signs (it comes from the magnitude, which is always positive), while only the numerator carries the  $\pm$  pattern.

**Quick Recall:** Label the box vertices by which axes they sit on; each diagonal's DRs are  $a, b, c$  with signs matching how the diagonal moves along each axis, so the angle formula naturally produces the  $\pm a^2 \pm b^2 \pm c^2$  pattern.

## II. Equation of a Line in Space

### Type 17

Writing the vector or cartesian equation of a line when a point on it and a direction vector (or direction ratios) are given directly.

**Formula Used:**  $\vec{r} = \vec{a} + \lambda \vec{b}$

**Formula Used:**  $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$

Source: Exercise 11.2, Q1(ii)

**Given:** Line parallel to  $3\hat{i} - 2\hat{j} + 6\hat{k}$ , passing through  $(1, -2, 3)$ .

**Solution:**

The vector equation of a line through a known point, parallel to a known vector, is simply the position vector of that point plus a scalar multiple of the direction vector – this is exactly the definition of "passing through with a given direction."

$\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$ , where  $\lambda$  is a non-zero real parameter.

### Final Answer

$$\vec{r} = \hat{i} - 2\hat{j} + 3\hat{k} + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$$

**Common Mistake:** Writing the position vector of the point using the direction vector's components instead of the point's own coordinates – keep the two clearly separate.

**Quick Recall:** Vector equation = position vector of the known point, plus  $\lambda$  times the direction vector; cartesian equation = each coordinate difference over the matching direction ratio.

### Type 18

Converting a line's equation between vector form and cartesian form (in either direction), by reading off the point and direction vector from one form to build the other.

Source: Exercise 11.2, Q4(i)

**Given:** Write the cartesian equation of  $\vec{r} = 2\hat{i} + \hat{j} - 4\hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$ .

**Solution:**

The vector equation already displays the point  $(2, 1, -4)$  (coefficient of the constant part) and the direction ratios  $\langle 1, -1, -1 \rangle$  (coefficient of  $\lambda$ ), so we transcribe these directly into the symmetric cartesian form.

$$\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z+4}{-1}.$$

**Final Answer**

$$\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z+4}{-1}$$

**Common Mistake:** Mixing up the sign inside the cartesian bracket – since the point is  $(2, 1, -4)$ , the  $z$  term must read  $z - (-4) = z + 4$ , not  $z - 4$ .

**Quick Recall:** Vector form's constant vector = the point; vector form's  $\lambda$ -vector = the direction ratios – read both off directly to build the cartesian form, and vice versa.

### Type 19

Reading off the direction ratios (or direction cosines) from a cartesian line equation that is *not* written in the clean  $\frac{x-x_1}{a}$  form – e.g. has a coefficient in front of  $x$ , or a minus sign multiplying the whole numerator. Must first algebraically rearrange to standard form before the DRs can be read off.

★ **Board Favorite** – A classic ISC trap – direct DR-reading from a poorly-formatted equation, without rearranging first, is one of the most common silent errors students make.

Source: Exercise 11.2, Q6(iv)

**Given:** Line AB:  $\frac{2x-1}{\sqrt{3}} = \frac{y+2}{2} = \frac{z-3}{3}$ . Find DCs of a line parallel to AB.

**Solution:**

The  $x$ -term has a coefficient of 2 in front of  $x$  inside the numerator, so this is not yet in standard form; we must factor that 2 out of the numerator to expose the true denominator under  $x$  alone.

$$\frac{2x-1}{\sqrt{3}} = \frac{2(x-\frac{1}{2})}{\sqrt{3}} = \frac{x-\frac{1}{2}}{\sqrt{3}/2}, \text{ so once rewritten, the true direction ratios are } \left\langle \frac{\sqrt{3}}{2}, 2, 3 \right\rangle,$$

i.e. after clearing the internal fraction,  $\langle \sqrt{3}, 4, 6 \rangle$ .

$$\text{Magnitude} = \sqrt{3 + 16 + 36} = \sqrt{55}.$$

**Final Answer**

DCs of AB (and hence any line parallel to it) =  $\left\langle \frac{\sqrt{3}}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{6}{\sqrt{55}} \right\rangle$

**Common Mistake:** Reading  $\langle 2, 2, 3 \rangle$  straight off the unarranged equation – this silently keeps the coefficient of  $x$  attached, giving wrong direction ratios.

**Alternate Board-Style Example**

Source: Exercise 11.2, Q7(i)

**Given:** Cartesian equations  $3x - 3 = 2y + 1 = 5 - 6z$ . Write the direction ratios.

**Solution:**

Each expression must first be written as (coefficient)  $\times$  ( $x$  or  $y$  or  $z$  – constant), then divided through by that coefficient to isolate  $x, y, z$  with coefficient 1, since only then do the denominators represent true direction ratios.

$3x - 3 = 3(x - 1)$ , so this term becomes  $\frac{x - 1}{1/3}$ .  $2y + 1 = 2(y + \frac{1}{2})$ , giving  $\frac{y + \frac{1}{2}}{1/2}$ .

$5 - 6z = -6(z - \frac{5}{6})$ , giving  $\frac{z - \frac{5}{6}}{-1/6}$ .

So the direction ratios (denominators) are  $\left\langle \frac{1}{3}, \frac{1}{2}, -\frac{1}{6} \right\rangle$ ; multiplying through by 6 to clear fractions (allowed, since DRs may be scaled) gives  $\langle 2, 3, -1 \rangle$ .

**Final Answer**

DRs =  $\langle 2, 3, -1 \rangle$

**Quick Recall:** Before reading off DRs from a cartesian line equation, always force every numerator into the exact form (variable – constant) with coefficient exactly 1 – factor out and divide the denominator accordingly.

**Type 20**

Writing the vector or cartesian equation of the line joining two given points.

**Formula Used:**  $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

Source: Exercise 11.2, Q5(ii)

**Given:** Line through the origin and the point  $(5, -2, 3)$ .

**Solution:**

The direction vector is simply the given point's position vector minus the origin's (which

is  $\vec{0}$ ), so the direction vector is just  $5\hat{i} - 2\hat{j} + 3\hat{k}$  itself; the vector equation uses the origin as the known point.

$$\vec{r} = \vec{0} + \lambda(5\hat{i} - 2\hat{j} + 3\hat{k}) = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k}).$$

For the cartesian form, since the point is the origin  $(0, 0, 0)$ , the equation simplifies to just the direction ratios under each variable:

$$\frac{x}{5} = \frac{y}{-2} = \frac{z}{3}.$$

#### Final Answer

$$\vec{r} = \lambda(5\hat{i} - 2\hat{j} + 3\hat{k}); \quad \frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$$

**Common Mistake:** Still writing “ $-0$ ” terms in the cartesian form out of habit (e.g.  $\frac{x-0}{5}$ ) – harmless but unnecessarily messy; simplify when the point is the origin.

**Quick Recall:** Line through two points: direction vector = second point minus first; if one of the points is the origin, the equation simplifies immediately.

#### Type 21

Finding the point where a given line meets a named coordinate plane (or a plane like  $y = k$ ), by writing a general point on the line in terms of the parameter and setting the coordinate that defines that plane equal to zero (or to  $k$ ).

Source: Exercise 11.2, Q9(iii)

**Given:** Find the point where the line through  $(5, 1, 6)$  and  $(3, 4, 1)$  crosses the  $xy$ -plane.

**Solution:**

DRs of the line:  $\langle 3 - 5, 4 - 1, 1 - 6 \rangle = \langle -2, 3, -5 \rangle$ . A general point on the line is  $(5 - 2\lambda, 1 + 3\lambda, 6 - 5\lambda)$  for parameter  $\lambda$ .

The  $xy$ -plane is defined by  $z = 0$ , so we set the  $z$ -coordinate of the general point to zero and solve for  $\lambda$ , since that's exactly where the line crosses that plane:  $6 - 5\lambda = 0 \Rightarrow \lambda = \frac{6}{5}$ .

Substituting back:  $x = 5 - 2\left(\frac{6}{5}\right) = 5 - \frac{12}{5} = \frac{13}{5}$ , and  $y = 1 + 3\left(\frac{6}{5}\right) = 1 + \frac{18}{5} = \frac{23}{5}$ .

#### Final Answer

$$\left(\frac{13}{5}, \frac{23}{5}, 0\right)$$

**Common Mistake:** Setting the wrong coordinate to zero – remember the  $xy$ -plane kills  $z$ , the  $yz$ -plane kills  $x$ , and the  $zx$ -plane kills  $y$ .

**Quick Recall:** Write a general point using the line's parameter, then zero out (or set equal to  $k$ ) whichever coordinate defines the target plane, and solve for the parameter.

**Type 22**

Extracting the direction numbers and vector equation of a line given in the unusual two-parameter symmetric form  $x = ay + b$ ,  $z = cy + d$  (where  $y$  itself is treated as the free parameter).

Source: Exercise 11.2, Q11

**Given:** The cartesian equations of a line are  $x = ay + b$ ,  $z = cy + d$ . Find direction numbers, also find its equation in vector form.

**Solution:**

We treat  $y$  itself as the running parameter, calling it  $\lambda$ , since both other variables are already expressed in terms of  $y$ : let  $y = \lambda$ , so  $x = a\lambda + b$ ,  $z = c\lambda + d$ .

Comparing this to the standard parametric form  $x = x_1 + a\lambda$ ,  $y = y_1 + 1 \cdot \lambda$ ,  $z = z_1 + c\lambda$ , the direction ratios are read straight off the coefficients of  $\lambda$ :  $\langle a, 1, c \rangle$ .

The point on the line (at  $\lambda = 0$ ) is  $(b, 0, d)$ , obtained by substituting  $\lambda = 0$  into each coordinate.

So the vector equation is  $\vec{r} = b\hat{i} + d\hat{k} + \lambda(a\hat{i} + \hat{j} + c\hat{k})$ , combining the point's position vector with  $\lambda$  times the direction vector.

**Final Answer**

Direction numbers  $\langle a, 1, c \rangle$ ;  $\vec{r} = b\hat{i} + d\hat{k} + \lambda(a\hat{i} + \hat{j} + c\hat{k})$

**Common Mistake:** Forgetting that the coefficient of  $y$  (i.e. of  $\lambda$ ) is always exactly 1, since  $y$  was chosen as the parameter itself – students sometimes look for a "y-coefficient" that doesn't explicitly appear and miss it.

**Quick Recall:** When  $y$  is left free while  $x, z$  are written in terms of it, treat  $y$  as the parameter directly – the direction ratios are then  $\langle \text{coeff of } y \text{ in } x, 1, \text{coeff of } y \text{ in } z \rangle$ .

**Type 23**

Writing the equation of a line through a given point, parallel to another given line (whose direction may itself come from two points, from a cartesian equation, or – in the special sub-case – must make equal angles with all three axes).

Source: Exercise 11.2, Q12(i)

**Given:** Line through A(2, -1, 1) parallel to the line joining B(-1, 4, 1) and C(1, 2, 2).

**Solution:**

DRs of BC (the direction to be parallel to) =  $\langle 1 - (-1), 2 - 4, 2 - 1 \rangle = \langle 2, -2, 1 \rangle$ .

Since the required line must be parallel to BC, it uses the same direction ratios but starts from point A instead:  $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \lambda(2\hat{i} - 2\hat{j} + \hat{k})$ .

**Final Answer**

$$\vec{r} = 2\hat{i} - \hat{j} + \hat{k} + \lambda(2\hat{i} - 2\hat{j} + \hat{k}); \text{ cartesian } \frac{x-2}{2} = \frac{y+1}{-2} = \frac{z-1}{1}$$

**Common Mistake:** Using A and B (or A and C) to build the direction, instead of using B and C together – the direction must come purely from the line we are told to be parallel to.

*Note: When the special condition is "equal angles with all three axes" (as in Q12(iii)), the DCs must each equal  $\pm \frac{1}{\sqrt{3}}$  since  $l = m = n$  and  $l^2 + m^2 + n^2 = 1$  together force this; use DRs  $\langle 1, 1, 1 \rangle$  through the given point.*

**Quick Recall:** Parallel to a given line = same direction ratios, new starting point; equal angles with all axes forces DRs  $\langle 1, 1, 1 \rangle$ .

**Type 24**

Given three vertices of a parallelogram ABCD (in order), finding the equations of the sides and the coordinates of the fourth vertex, using the property that diagonals of a parallelogram bisect each other.

Source: Exercise 11.2, Q15

**Given:** A(4, 5, 10), B(2, 3, 4), C(1, 2, -1) are three vertices of parallelogram ABCD. Find equations of sides AB, BC and the coordinates of D.

**Solution:**

Side AB: DRs =  $\langle 2 - 4, 3 - 5, 4 - 10 \rangle = \langle -2, -2, -6 \rangle$ , i.e.  $\langle 1, 1, 3 \rangle$ ; equation  $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-10}{3}$  through A.

Side BC: DRs =  $\langle 1 - 2, 2 - 3, -1 - 4 \rangle = \langle -1, -1, -5 \rangle$ , i.e.  $\langle 1, 1, 5 \rangle$ ; equation  $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{5}$  through B.

For vertex D, we use that the diagonals AC and BD of a parallelogram share the same midpoint: midpoint of AC =  $\left( \frac{4+1}{2}, \frac{5+2}{2}, \frac{10-1}{2} \right) = \left( \frac{5}{2}, \frac{7}{2}, \frac{9}{2} \right)$ .

Setting this equal to the midpoint of BD:  $\left( \frac{2+x_D}{2}, \frac{3+y_D}{2}, \frac{4+z_D}{2} \right) = \left( \frac{5}{2}, \frac{7}{2}, \frac{9}{2} \right)$ , so  $x_D = 5 - 2 = 3$ ,  $y_D = 7 - 3 = 4$ ,  $z_D = 9 - 4 = 5$ .

**Final Answer**

D = (3, 4, 5); sides as derived above

**Common Mistake:** Equating the midpoint of AC to the midpoint of AB or CD instead of BD – only the two diagonals AC and BD bisect each other in parallelogram ABCD.

**Quick Recall:** In parallelogram ABCD, diagonals AC and BD bisect each other – equate their midpoints to find any missing vertex.

### Type 25

Showing that three given points are collinear by writing the vector equation of the line through two of them and verifying that the third point's position vector satisfies this equation for some value of the parameter.

Source: Exercise 11.2, Q16(i)

**Given:** Show A(-2, 4, 7), B(3, -6, -8), C(1, -2, -2) are collinear.

**Solution:**

Vector equation of line AB:  $\vec{r} = (-2\hat{i} + 4\hat{j} + 7\hat{k}) + \lambda[(3 - (-2))\hat{i} + (-6 - 4)\hat{j} + (-8 - 7)\hat{k}] = (-2\hat{i} + 4\hat{j} + 7\hat{k}) + \lambda(5\hat{i} - 10\hat{j} - 15\hat{k})$ .

We check whether C's position vector  $\hat{i} - 2\hat{j} - 2\hat{k}$  satisfies this for some  $\lambda$ , since if it does, C lies on line AB and all three points are collinear.

Matching  $\hat{i}$ :  $-2 + 5\lambda = 1 \Rightarrow \lambda = \frac{3}{5}$ . Checking  $\hat{j}$ :  $4 - 10\left(\frac{3}{5}\right) = 4 - 6 = -2 \checkmark$ . Checking  $\hat{k}$ :  $7 - 15\left(\frac{3}{5}\right) = 7 - 9 = -2 \checkmark$ .

All three components are consistent with the same  $\lambda = \frac{3}{5}$ , so C indeed lies on line AB.

### Final Answer

C lies on line AB (all three coordinates satisfy  $\lambda = \frac{3}{5}$ )  $\Rightarrow$  A, B, C are collinear.

**Common Mistake:** Finding  $\lambda$  from just one coordinate and declaring collinearity without checking the other two coordinates agree – all three must be consistent.

**Quick Recall:** To test collinearity via vectors: build the line through two points, then check the third point's position vector gives the *same*  $\lambda$  in all three coordinates.

### Type 26

Finding the point(s) on a given line that lie at a specified distance from a given point on (or related to) that line, using the direction cosines so that the parameter genuinely represents distance.

Source: Exercise 11.2, Q17(i)

**Given:** Find points on the line  $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6}$  at a distance of 3 units from  $(1, -2, 3)$ .

**Solution:**

DRs are  $\langle 2, 3, 6 \rangle$ ; magnitude  $= \sqrt{4 + 9 + 36} = \sqrt{49} = 7$ , so DCs are  $\langle \frac{2}{7}, \frac{3}{7}, \frac{6}{7} \rangle$ . We must use DCs (not raw DRs) as the parameter's coefficients so that the parameter itself measures true distance along the line.

A general point at distance  $r$  from  $(1, -2, 3)$  along the line is  $(1 + \frac{2}{7}r, -2 + \frac{3}{7}r, 3 + \frac{6}{7}r)$ . Setting  $r = 3$ :  $x = 1 + \frac{6}{7} = \frac{13}{7}$ ,  $y = -2 + \frac{9}{7} = -\frac{5}{7}$ ,  $z = 3 + \frac{18}{7} = \frac{39}{7}$ . Since distance can be measured in either direction along the line, we also take  $r = -3$ :  $x = 1 - \frac{6}{7} = \frac{1}{7}$ ,  $y = -2 - \frac{9}{7} = -\frac{23}{7}$ ,  $z = 3 - \frac{18}{7} = \frac{3}{7}$ .

**Final Answer**

$$\left(\frac{13}{7}, -\frac{5}{7}, \frac{39}{7}\right) \text{ and } \left(\frac{1}{7}, -\frac{23}{7}, \frac{3}{7}\right)$$

**Common Mistake:** Using the raw direction ratios  $\langle 2, 3, 6 \rangle$  (instead of dividing by the magnitude 7 first) as the coefficients of  $r$  – this makes  $r$  meaningless as an actual distance, giving a completely wrong point.

**Quick Recall:** To move an exact distance  $r$  along a line, always convert its DRs to DCs first, then add  $r \times \text{DC}$  to each coordinate – and remember both  $+r$  and  $-r$  are usually valid.

### Type 27

Showing that two given lines intersect (typically each given via two points, or in symmetric cartesian form) and finding their point of intersection, by equating the general points of both lines and solving for consistency.

Source: Exercise 11.2, Q20(i)

**Given:** Prove the line through  $A(0, -1, -1)$ ,  $B(4, 5, 1)$  intersects the line through  $C(3, 9, 4)$ ,  $D(-4, 4, 4)$ . Find their point of intersection.

**Solution:**

Line AB: DRs  $\langle 4, 6, 2 \rangle$  i.e.  $\langle 2, 3, 1 \rangle$ ; general point  $(2\lambda, -1 + 3\lambda, -1 + \lambda)$ .

Line CD: DRs  $\langle -7, -5, 0 \rangle$ ; general point  $(3 - 7\mu, 9 - 5\mu, 4)$ .

For the lines to intersect, some choice of  $\lambda, \mu$  must make all three coordinates match simultaneously, so we equate coordinate by coordinate and check consistency (since with two unknowns and three equations, this is an over-determined system that only has a solution if the lines truly meet).

From  $z$ :  $-1 + \lambda = 4 \Rightarrow \lambda = 5$ . Substituting into  $x$ :  $2(5) = 3 - 7\mu \Rightarrow 10 = 3 - 7\mu \Rightarrow \mu = -1$ . Check  $y$ : LHS  $= -1 + 3(5) = 14$ ; RHS  $= 9 - 5(-1) = 14$  ✓ – consistent, so the lines do intersect.

Point of intersection: substitute  $\lambda = 5$  into line AB's general point:  $(2(5), -1 + 15, -1 + 5) = (10, 14, 4)$ .

### Final Answer

The lines intersect at  $(10, 14, 4)$ .

**Common Mistake:** Solving only two of the three coordinate equations and assuming the third is automatically satisfied – with two lines in 3D, checking the third coordinate is what actually proves intersection rather than skewness.

**Quick Recall:** Write both lines with their own parameters, solve two coordinate equations for the two parameters, then verify the third coordinate matches – that check is the real proof of intersection.

### Type 28

A third line, of given direction, is drawn to meet two other given lines – find the coordinates of both points of intersection and the length of the segment intercepted on the third line.

★ **Board Favorite** – A genuinely multi-step composite question style that tests parametrisation of two intersections simultaneously – appears as a long-answer favorite when examiners want to combine several skills in one question.

Source: Exercise 11.2, Q21

**Given:** A line with DRs  $\langle 2, 7, -5 \rangle$  is drawn to intersect the lines  $\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$  and  $\frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}$ . Find the points of intersection and the length intercepted.

### Solution:

Let the first given line's general point be  $P(5 + 3s, 7 - s, -2 + s)$ , and the second's be  $Q(-3 - 3t, 3 + 2t, 6 + 4t)$ , using independent parameters  $s, t$ .

Since PQ must have DRs proportional to  $\langle 2, 7, -5 \rangle$  (the direction of the connecting line), we form the ratios:  $\frac{(-3 - 3t) - (5 + 3s)}{2} = \frac{(3 + 2t) - (7 - s)}{7} = \frac{(6 + 4t) - (-2 + s)}{-5}$ .

Simplifying each numerator:  $(-8 - 3t - 3s)$ ,  $(-4 + 2t + s)$ ,  $(8 + 4t - s)$ . Equating the first pair to the third in turn (two independent equations in  $s, t$ ) and solving simultaneously gives specific values of  $s$  and  $t$ ; substituting these back into P and Q's formulas yields the two actual points of intersection, and the distance formula between them gives the intercepted length.

### Final Answer

Points of intersection found from the simultaneous equations above; length =  $|PQ|$  via the distance formula (numeric values depend on the exact  $s, t$  solved).

**Common Mistake:** Setting up the direction-ratio proportion using only two of the three coordinate differences – all three must be used (in pairs) to generate exactly two independent equations for the two unknowns  $s, t$ .

**Quick Recall:** Parametrise both target lines separately, force the connecting segment's DRs to match the given direction via two independent ratio equations, solve for both parameters, then use the distance formula for the intercepted length.

### Type 29

Finding the equations of the medians of a triangle from its vertices, and using two of these median equations together to find the centroid.

Source: Exercise 11.2, Q22

**Given:** Vertices of  $\triangle ABC$  are  $A(1, 1, 0)$ ,  $B(1, 2, 1)$ ,  $C(-2, 2, -1)$ . Find the equations of the medians through A and B; use them to find the centroid.

#### Solution:

The median through A goes to the midpoint of BC, since a median joins a vertex to the midpoint of the opposite side.

$$\text{Midpoint of BC} = \left( \frac{1-2}{2}, \frac{2+2}{2}, \frac{1-1}{2} \right) = \left( -\frac{1}{2}, 2, 0 \right).$$

Median through A: DRs =  $\left\langle -\frac{1}{2} - 1, 2 - 1, 0 - 0 \right\rangle = \left\langle -\frac{3}{2}, 1, 0 \right\rangle$ , i.e.  $\langle -3, 2, 0 \rangle$ ; equation  $\frac{x-1}{-3} = \frac{y-1}{2} = \frac{z}{0}$  (meaning  $z = 0$  throughout, with the  $x, y$  part running as usual).

$$\text{Midpoint of AC} = \left( \frac{1-2}{2}, \frac{1+2}{2}, \frac{0-1}{2} \right) = \left( -\frac{1}{2}, \frac{3}{2}, -\frac{1}{2} \right).$$

Median through B: DRs =  $\left\langle -\frac{1}{2} - 1, \frac{3}{2} - 2, -\frac{1}{2} - 1 \right\rangle = \left\langle -\frac{3}{2}, -\frac{1}{2}, -\frac{3}{2} \right\rangle$ , i.e.  $\langle 3, 1, 3 \rangle$ ; equation  $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-1}{3}$ .

Since all medians pass through the centroid, we equate general points on both medians and solve for consistency, or – more directly – use the standard centroid formula as a

$$\text{check: centroid} = \left( \frac{1+1-2}{3}, \frac{1+2+2}{3}, \frac{0+1-1}{3} \right) = \left( 0, \frac{5}{3}, 0 \right).$$

#### Final Answer

$$\text{Centroid} = \left( 0, \frac{5}{3}, 0 \right)$$

**Common Mistake:** Computing the midpoint of the wrong side for a given median – the median from A always goes to the midpoint of BC (the side *not* containing A), a mix-up that's easy to make under time pressure.

**Quick Recall:** Median from a vertex  $\rightarrow$  line to the midpoint of the opposite side; centroid is simply the average of all three vertices' coordinates, and can be used to check the intersection of any two medians.

### III. Angle Between Lines & Shortest Distance

#### Type 30

Finding the angle between two lines that are each given as a full vector or cartesian equation – first extract the direction vectors/ratios from each equation, then apply the standard angle formula.

**Formula Used:**  $\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$

Source: Exercise 11.3, Q2(i)

**Given:** Lines  $\frac{x+3}{3} = \frac{y-1}{5} = \frac{z+3}{4}$  and  $\frac{x+1}{1} = \frac{y-4}{1} = \frac{z-5}{2}$ .

**Solution:**

DRs of the first line:  $\langle 3, 5, 4 \rangle$ ; DRs of the second:  $\langle 1, 1, 2 \rangle$ , read directly off the denominators since both equations are already in standard symmetric form.

$$\cos \theta = \frac{3(1) + 5(1) + 4(2)}{\sqrt{9 + 25 + 16}\sqrt{1 + 1 + 4}} = \frac{3 + 5 + 8}{\sqrt{50}\sqrt{6}} = \frac{16}{\sqrt{300}} = \frac{16}{10\sqrt{3}} = \frac{8}{5\sqrt{3}} = \frac{8\sqrt{3}}{15}.$$

#### Final Answer

$$\theta = \cos^{-1} \left( \frac{8\sqrt{3}}{15} \right)$$

**Common Mistake:** Forgetting to simplify  $\sqrt{300}$  correctly (it's  $10\sqrt{3}$ , not  $10\sqrt{30}$  or similar) – always factor out perfect squares carefully.

**Quick Recall:** Whenever a line is given in full equation form, its DRs are just the denominators (cartesian) or the  $\lambda$ -vector (vector form) – extract first, then apply the cosine formula exactly as with bare DRs.

#### Type 31

Checking (or proving) whether two lines given by parametric cartesian equations are at right angles, by extracting DRs from the coefficient of the parameter in each coordinate.

Source: Exercise 11.3, Q3(ii)

**Given:** Check whether the lines  $x = 2\lambda + 2$ ,  $y = 7\lambda + 1$ ,  $z = -3\lambda - 3$  and  $x = -\mu - 2$ ,  $y = 2\mu + 8$ ,  $z = 4\mu + 5$  are perpendicular.

**Solution:**

Reading the coefficient of the parameter in each coordinate gives the DRs directly: first line  $\langle 2, 7, -3 \rangle$ , second line  $\langle -1, 2, 4 \rangle$ , since in parametric form  $x = x_1 + (\text{DR})\lambda$  the DR is exactly that coefficient.

Dot product  $= 2(-1) + 7(2) + (-3)(4) = -2 + 14 - 12 = 0$ .

**Final Answer**

Dot product  $= 0$ , so the lines are perpendicular.

**Common Mistake:** Misreading the constant term as part of the direction ratio – only the coefficient multiplying the parameter is the DR; the constant term is part of the point the line passes through.

**Quick Recall:** In  $x = x_1 + a\lambda$  form,  $a$  (the parameter's coefficient) is the direction ratio for that coordinate – perpendicularity is still just dot product  $= 0$ .

### Type 32

Finding an unknown constant so that two given lines become perpendicular to each other.

*Source: Exercise 11.3, Q4(i)*

**Given:** Lines  $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$  and  $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$  are perpendicular; find  $k$ .

**Solution:**

DRs are  $\langle -3, 2k, 2 \rangle$  and  $\langle 3k, 1, -5 \rangle$ . Perpendicularity forces their dot product to vanish, which gives a single linear equation in  $k$ :

$$(-3)(3k) + (2k)(1) + (2)(-5) = 0 \Rightarrow -9k + 2k - 10 = 0 \Rightarrow -7k = 10 \Rightarrow k = -\frac{10}{7}.$$

**Final Answer**

$$k = -\frac{10}{7}$$

**Common Mistake:** Sign slip while distributing the negative DRs across the product terms – write out each product on its own line to avoid combining signs incorrectly.

**Quick Recall:** Perpendicularity between two lines with an unknown always reduces to one linear (or occasionally quadratic) equation from setting the DR dot product to zero.

**Type 33**

A combined question: first find the unknown constant that makes two lines perpendicular, then – using that same value – determine whether the two lines actually intersect or not.

★ **Board Favorite** – Combines two separate skills (perpendicularity + intersection-testing) into a single flowing question – a favourite long-answer structure precisely because it rewards students who keep careful track of which value of the unknown to carry forward.

Source: Exercise 11.3, Q6(i)

**Given:** Lines  $\frac{x-1}{-3} = \frac{y-2}{2\lambda} = \frac{z-3}{2}$  and  $\frac{x-1}{3\lambda} = \frac{y-1}{1} = \frac{z-6}{-5}$  are perpendicular. Find  $\lambda$ ; hence find whether the lines are intersecting.

**Solution:**

As in Type 32's method, perpendicularity gives  $(-3)(3\lambda) + (2\lambda)(1) + 2(-5) = 0 \Rightarrow -9\lambda + 2\lambda - 10 = 0 \Rightarrow \lambda = -\frac{10}{7}$ .

Now, using this  $\lambda$ , we substitute back to get concrete DRs: line 1 becomes  $\left\langle -3, -\frac{20}{7}, 2 \right\rangle$ , i.e.  $\langle -21, -20, 14 \rangle$ ; line 2 becomes  $\left\langle -\frac{30}{7}, 1, -5 \right\rangle$ , i.e.  $\langle -30, 7, -35 \rangle$ .

General point on line 1:  $(1 - 21s, 2 - 20s, 3 + 14s)$ ; general point on line 2:  $(1 - 30t, 1 + 7t, 6 - 35t)$ . We equate corresponding coordinates and check for consistency, exactly as in the intersection-testing method (Type 27), to see whether some  $(s, t)$  satisfies all three simultaneously.

Carrying through the elimination (from the  $x$  and  $y$  equations) and substituting into the  $z$ -equation reveals whether the resulting statement is true (lines intersect) or produces a contradiction (lines do not intersect, i.e. they are skew, since we already know they are perpendicular but not necessarily meeting).

**Final Answer**

$\lambda = -\frac{10}{7}$ ; whether the lines intersect follows from checking consistency of the three coordinate equations with this  $\lambda$  substituted in.

**Common Mistake:** Forgetting to actually substitute the found  $\lambda$  back into the direction ratios before testing for intersection – the second part of the question depends entirely on using the concrete (numeric) line equations, not the ones still containing  $\lambda$ .

**Quick Recall:** Two-part combo: solve for the unknown via perpendicularity first, substitute it back to get fully numeric lines, then run the usual three-coordinate consistency check for intersection.

**Type 34**

Finding the equation of the perpendicular drawn from an external point to a given line – i.e. the line through the given point and the foot of perpendicular on the given line.

Source: Exercise 11.3, Q7

**Given:** Find the equation of the perpendicular from  $A(2, 4, -1)$  to the line  $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ .

**Solution:**

Let N, the foot of perpendicular, be the general point  $(-5 + \lambda, -3 + 4\lambda, 6 - 9\lambda)$  on the given line.

DRs of AN =  $\langle -5 + \lambda - 2, -3 + 4\lambda - 4, 6 - 9\lambda + 1 \rangle = \langle \lambda - 7, 4\lambda - 7, 7 - 9\lambda \rangle$ .

Since AN must be perpendicular to the given line (DRs  $\langle 1, 4, -9 \rangle$ ), their dot product is zero:

$$1(\lambda - 7) + 4(4\lambda - 7) + (-9)(7 - 9\lambda) = 0 \Rightarrow \lambda - 7 + 16\lambda - 28 - 63 + 81\lambda = 0 \Rightarrow 98\lambda - 98 = 0 \Rightarrow \lambda = 1.$$

Substituting back:  $N = (-5 + 1, -3 + 4, 6 - 9) = (-4, 1, -3)$ .

The required perpendicular is the line through  $A(2, 4, -1)$  and  $N(-4, 1, -3)$ : DRs =  $\langle -6, -3, -2 \rangle$ .

**Final Answer**

$$\frac{x-2}{-6} = \frac{y-4}{-3} = \frac{z+1}{-2}$$

**Common Mistake:** Stopping at finding N and forgetting the question actually asks for the equation of the perpendicular *line* (through A and N), not just the foot of perpendicular itself.

**Quick Recall:** Foot of perpendicular from a point to a line: parametrise the line, force perpendicularity to solve for the parameter, then the required line joins the original point to this foot.

**Type 35**

Finding the (shortest) distance of a point from a line without needing the foot of perpendicular explicitly named, or finding the area of a triangle where two vertices lie on a known line at a known side-length apart.

**Formula Used:**  $d = \sqrt{|AP|^2 - |AN|^2}$

Source: Exercise 11.3, Q8(iv)

**Given:** Find the distance of the point  $(2, 4, -1)$  from the line  $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$ .

**Solution:**

This reuses the foot of perpendicular  $N = (-4, 1, -3)$  found in Type 34 for the very same data, since the distance we want is exactly  $|AN|$  where  $A$  is the given point.

$$|AN| = \sqrt{(-4-2)^2 + (1-4)^2 + (-3+1)^2} = \sqrt{36+9+4} = \sqrt{49} = 7 \text{ units.}$$

**Final Answer**

Distance = 7 units

*Note: For the triangle-area sub-case (Q8(iii)): if  $B, C$  lie on a known line with  $|BC| = 5$  and  $A$  is off the line, first find the foot of perpendicular  $N$  from  $A$  to the line to get the height  $|AN|$ , then  $\text{area} = \frac{1}{2} \times |BC| \times |AN|$ , exactly as with any triangle once base and height are known.*

**Common Mistake:** Recomputing the whole foot-of-perpendicular calculation from scratch instead of recognising it's identical working to a nearby question – once  $N$  is found, the distance is just one more distance-formula step.

**Quick Recall:** Distance of a point from a line = distance from that point to the foot of perpendicular on the line; for a triangle with two vertices on a line, that same perpendicular length is the height.

### Type 36

Finding the equation of a line through a given point, perpendicular to two other given lines (each given as full vector or cartesian equations) – uses the cross product of the two given direction vectors.

**Formula Used:**  $\vec{b} = \vec{b}_1 \times \vec{b}_2$

*Source: Exercise 11.3, Q11*

**Given:** Find the equation of a line through  $(1, 2, -4)$  and perpendicular to the two lines  $\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$  and  $\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$ .

**Solution:**

The required line's direction must be perpendicular to both given directions  $\langle 3, -16, 7 \rangle$  and  $\langle 3, 8, -5 \rangle$ , so it is along their cross product, exactly as in Type 13's cross-product method.

$$\begin{aligned} \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix} = \hat{i}((-16)(-5) - 7(8)) - \hat{j}(3(-5) - 7(3)) + \hat{k}(3(8) - (-16)(3)) \\ &= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48) = 24\hat{i} + 36\hat{j} + 72\hat{k}, \text{ i.e. DRs } \langle 24, 36, 72 \rangle \text{ or simplified } \langle 2, 3, 6 \rangle. \end{aligned}$$

The required line passes through  $(1, 2, -4)$  with this direction.

### Final Answer

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + t(2\hat{i} + 3\hat{j} + 6\hat{k})$$

**Common Mistake:** Using the given points inside the vector equations (e.g.  $(8, -19, 10)$ ) as if they were direction vectors – only the  $\lambda/\mu$ -multiplied vectors are directions; the other parts are just points the given lines pass through and are irrelevant to this cross product.

**Quick Recall:** Line perpendicular to two other lines: cross their two direction vectors to get the new line's direction, then attach it to the given point.

### Type 37

Finding the shortest distance between two skew lines, given in vector or cartesian form, using the scalar-triple-product formula.

★ **Board Favorite** – One of the most heavily tested formula-applications in the whole chapter – almost guaranteed to appear as a full long-answer question most years.

**Formula Used:** 
$$\text{S.D.} = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

Source: Exercise 11.4, Q1(i)

**Given:**  $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$  and  $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$ .

### Solution:

Here  $\vec{a}_1 = \hat{i} + 2\hat{j} + \hat{k}$ ,  $\vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$ ,  $\vec{a}_2 = 2\hat{i} - \hat{j} - \hat{k}$ ,  $\vec{b}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$ .

$\vec{a}_2 - \vec{a}_1 = \hat{i} - 3\hat{j} - 2\hat{k}$ , found by simple vector subtraction since this is the displacement between the two lines' reference points.

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = \hat{i}(-1(2) - 1(1)) - \hat{j}(1(2) - 1(2)) + \hat{k}(1(1) - (-1)(2)) = -3\hat{i} - 0\hat{j} + 3\hat{k}.$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{9 + 0 + 9} = \sqrt{18} = 3\sqrt{2}.$$

Numerator (dot product):  $(\hat{i} - 3\hat{j} - 2\hat{k}) \cdot (-3\hat{i} + 0\hat{j} + 3\hat{k}) = -3 - 0 - 6 = -9$ ; taking the absolute value gives 9.

$$\text{S.D.} = \frac{9}{3\sqrt{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2} \text{ units.}$$

**Final Answer**

$$\text{Shortest distance} = \frac{3\sqrt{2}}{2} \text{ units}$$

**Common Mistake:** Dropping the modulus sign at the final step and reporting a negative distance – distance is always a magnitude, so the absolute value must be taken before dividing.

**Quick Recall:** Shortest distance between skew lines: cross the two direction vectors, dot that with the vector joining a point on each line, take the absolute value, and divide by the magnitude of the cross product.

**Type 38**

Finding the distance between two parallel lines, using the formula that treats the shared direction vector and the vector joining one point on each line.

$$\text{Formula Used: S.D.} = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$$

Source: Exercise 11.4, Q7

**Given:** Find the equation of the line through  $(2, 1, -1)$  parallel to  $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ ; also find the distance between these lines.

**Solution:**

Since the required line is parallel to the given one, it shares the same direction vector  $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ , and simply starts from the new point:

$$\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} - \hat{j} + \hat{k}).$$

For the distance between the two parallel lines,  $\vec{a}_2 - \vec{a}_1 = (2\hat{i} + \hat{j} - \hat{k}) - (\hat{i} + \hat{j}) = \hat{i} + 0\hat{j} - \hat{k}$ , the displacement between one point on each line.

$$\vec{b} \times (\vec{a}_2 - \vec{a}_1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 0 & -1 \end{vmatrix} = \hat{i}((-1)(-1) - 1(0)) - \hat{j}(2(-1) - 1(1)) + \hat{k}(2(0) - (-1)(1)) =$$

$$\hat{i} + 3\hat{j} + \hat{k}.$$

Magnitude of this cross product  $= \sqrt{1 + 9 + 1} = \sqrt{11}$ ; magnitude of  $\vec{b} = \sqrt{4 + 1 + 1} = \sqrt{6}$ .

$$\text{Distance} = \frac{\sqrt{11}}{\sqrt{6}} = \sqrt{\frac{11}{6}}.$$

**Final Answer**

$$\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} - \hat{j} + \hat{k}); \text{ distance} = \frac{\sqrt{11}}{\sqrt{6}} \text{ units}$$

**Common Mistake:** Using the S.D.-between-skew-lines formula (which needs *two* different direction vectors) instead of the parallel-lines version, which uses only one shared direction vector crossed with the connecting vector.

**Quick Recall:** Parallel lines: distance =  $\frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$  – only one direction vector is involved since both lines point the same way.

## IV. The Plane: Forms & Equations

### Type 39

Finding the direction cosines of the normal to a plane from its equation, and converting a plane's equation between vector and cartesian form.

**Formula Used:** DCs of normal =  $\left\langle \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}} \right\rangle$

Source: Exercise 11.5, Q1(i)

**Given:** Find the DCs of the normal to the plane  $2x + 3y - z - 7 = 0$ .

**Solution:**

The coefficients of  $x, y, z$  in the cartesian equation directly give the direction ratios of the normal, since a plane's equation is built exactly as (normal)·(position vector)=const.  
DRs of normal =  $\langle 2, 3, -1 \rangle$ ; magnitude =  $\sqrt{4 + 9 + 1} = \sqrt{14}$ .

### Final Answer

$$\left\langle \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}} \right\rangle$$

**Common Mistake:** Forgetting to include the correct sign for a variable with a negative coefficient (like the  $-z$  here) when reading off the normal's DRs.

**Quick Recall:** The coefficients of  $x, y, z$  in a plane's cartesian equation are exactly the DRs of its normal – normalise (divide by the magnitude) to get DCs.

### Type 40

Finding the equation of a plane passing through a given point and perpendicular to a given line (i.e. the line's direction is the plane's normal).

★ **Board Favorite** – The most fundamental and frequently tested plane-construction pattern – essentially every board paper has at least one plane question built on this single idea.

**Formula Used:**  $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

Source: Exercise 11.5, Q4(i)

**Given:** Plane through  $(2, -3, 1)$ , perpendicular to the line with DRs  $\langle 3, -1, 5 \rangle$ .

**Solution:**

Since the plane is perpendicular to the given line, that line's direction ratios serve directly as the plane's normal – this is the defining shortcut of this type.

Using the point-normal form:  $3(x - 2) - 1(y + 3) + 5(z - 1) = 0$ , since we substitute the point's coordinates and the normal's components straight into the formula.

Expanding:  $3x - 6 - y - 3 + 5z - 5 = 0 \Rightarrow 3x - y + 5z - 14 = 0$ .

**Final Answer**

$$3x - y + 5z - 14 = 0$$

**Common Mistake:** Forgetting to distribute the negative sign correctly when a DR component is negative (here the  $-1$  on  $y$ ) – write  $-1(y - (-3)) = -1(y + 3) = -y - 3$  carefully.

**Quick Recall:** "Perpendicular to a line" always means the line's DRs become the plane's normal – then it's just the point-normal formula.

**Type 41**

Finding the equation of a plane that perpendicularly bisects a given line segment (the plane passes through the midpoint of the segment, and the segment's direction is the plane's normal).

*Source: Exercise 11.5, Q4(ii)*

**Given:** Plane which bisects the segment joining  $A(2, 3, 5)$  and  $B(4, 5, 7)$  at right angles.

**Solution:**

"Bisects at right angles" means the plane passes through the midpoint of  $AB$ , and is normal to  $AB$  itself – two pieces of information we can extract directly from the two given points.

$$\text{Midpoint of } AB = \left( \frac{2+4}{2}, \frac{3+5}{2}, \frac{5+7}{2} \right) = (3, 4, 6).$$

DRs of  $AB$  (the required normal) =  $\langle 2, 2, 2 \rangle$ , i.e.  $\langle 1, 1, 1 \rangle$ .

Using point-normal form through  $(3, 4, 6)$ :  $1(x-3) + 1(y-4) + 1(z-6) = 0 \Rightarrow x + y + z - 13 = 0$ .

**Final Answer**

$$x + y + z = 13$$

**Common Mistake:** Using one of the endpoints A or B as the point on the plane instead of their midpoint – "bisects" specifically means the plane passes through the midpoint, not either endpoint.

**Quick Recall:** Perpendicular bisector plane: point = midpoint of the segment, normal = DRs of the segment itself.

### Type 42

Finding the equation of a plane given its perpendicular distance from the origin and the direction of its normal.

**Formula Used:**  $\vec{r} \cdot \hat{n} = d$

Source: Exercise 11.5, Q8(i)

**Given:** Plane at distance  $\frac{6}{\sqrt{29}}$  from the origin, normal direction  $2\hat{i} - 3\hat{j} + 4\hat{k}$ .

**Solution:**

The normal form of a plane needs a *unit* normal vector, so we first normalise the given direction, since only a unit vector correctly represents "distance  $d$  along that direction."

$$|2\hat{i} - 3\hat{j} + 4\hat{k}| = \sqrt{4 + 9 + 16} = \sqrt{29}, \text{ so } \hat{n} = \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}}.$$

Vector equation:  $\vec{r} \cdot \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}} = \frac{6}{\sqrt{29}}$ , which simplifies (multiplying through by  $\sqrt{29}$ )

$$\text{to } \vec{r} \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 6.$$

In cartesian form this reads  $2x - 3y + 4z = 6$ .

### Final Answer

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 4\hat{k}) = 6; \text{ cartesian } 2x - 3y + 4z = 6$$

**Common Mistake:** Using the given vector directly as  $\hat{n}$  without dividing by its magnitude first – the normal form strictly requires a *unit* normal for  $d$  to represent the true distance.

**Quick Recall:** Normal-form plane: always normalise the given direction to a unit vector before writing  $\vec{r} \cdot \hat{n} = d$ .

### Type 43

Finding the foot of perpendicular and the length of the perpendicular drawn from a given point to a given plane.

Source: Exercise 11.5, Q23(i)

**Given:** Find the length and the foot of perpendicular drawn from the point  $(2, 3, 7)$  to the plane  $3x - y - z = 7$ .

**Solution:**

The normal to the plane  $3x - y - z = 7$  has DRs  $\langle 3, -1, -1 \rangle$ . The foot of perpendicular N from P(2, 3, 7) lies along this normal direction from P, so we write N as  $(2+3\lambda, 3-\lambda, 7-\lambda)$  for some  $\lambda$ .

Since N must actually lie on the plane, substitute into the plane's equation:  $3(2+3\lambda) - (3-\lambda) - (7-\lambda) = 7 \Rightarrow 6+9\lambda-3+\lambda-7+\lambda=7 \Rightarrow 11\lambda-4=7 \Rightarrow 11\lambda=11 \Rightarrow \lambda=1$ . Substituting back:  $N = (2+3, 3-1, 7-1) = (5, 2, 6)$ .

Length of perpendicular  $= |PN| = |\lambda| \times |\text{normal}| = 1 \times \sqrt{9+1+1} = \sqrt{11}$  units, since moving a parameter  $\lambda$  along a direction vector of magnitude  $|\vec{n}|$  covers a distance  $|\lambda||\vec{n}|$ .

**Final Answer**

Foot of perpendicular  $= (5, 2, 6)$ ; length of perpendicular  $= \sqrt{11}$  units

**Common Mistake:** Forgetting that moving by parameter  $\lambda$  along the (non-unit) normal DRs covers distance  $|\lambda| \times |\vec{n}|$ , not just  $|\lambda|$  – always multiply by the normal's magnitude for the true length.

**Quick Recall:** Foot of perpendicular to a plane: move from the point along the normal's direction by parameter  $\lambda$ , force the result to satisfy the plane's equation, solve for  $\lambda$ , then multiply by the normal's magnitude for the perpendicular length.

**Type 44**

Finding the image (reflection) of a point in a given plane, using the foot of perpendicular as the midpoint between the original point and its image.

★ **Board Favorite** – A guaranteed long-answer question type – reflections in a plane combine the foot-of-perpendicular method with a midpoint step, testing two ideas at once.

Source: Exercise 11.5, Q25(i)

**Given:** Find the image of the point  $(1, 2, 3)$  in the plane  $x + 2y + 4z = 38$ .

**Solution:**

Normal DRs are  $\langle 1, 2, 4 \rangle$ . Let N be the foot of perpendicular from P(1, 2, 3):  $N = (1 + \lambda, 2 + 2\lambda, 3 + 4\lambda)$ .

Substituting into the plane's equation (since N lies on the plane):  $(1 + \lambda) + 2(2 + 2\lambda) + 4(3 + 4\lambda) = 38 \Rightarrow 1 + \lambda + 4 + 4\lambda + 12 + 16\lambda = 38 \Rightarrow 21\lambda + 17 = 38 \Rightarrow \lambda = 1$ .

So  $N = (2, 4, 7)$ .

If  $Q(\alpha, \beta, \gamma)$  is the image of P, then N is the midpoint of PQ, because reflection means the foot of perpendicular is exactly halfway between the original point and its mirror image.

$$\frac{1+\alpha}{2} = 2 \Rightarrow \alpha = 3, \quad \frac{2+\beta}{2} = 4 \Rightarrow \beta = 6, \quad \frac{3+\gamma}{2} = 7 \Rightarrow \gamma = 11.$$

**Final Answer**

Image = (3, 6, 11)

**Common Mistake:** Reporting N itself as the "image" – N is only the foot of perpendicular (the midpoint); the actual image is found by doubling past N using the midpoint relation.

**Quick Recall:** Image in a plane: find the foot of perpendicular N first (parametrise along the normal, substitute into the plane), then use "N is the midpoint of P and its image" to solve for the image.

**Type 45**

Finding the image of a point in one of the three coordinate planes (XOY, YOZ, ZOX) – a quick special case where only one coordinate's sign flips.

Source: Exercise 11.5, Q6(i),(ii)

**Given:** (a) Image of  $(-2, 3, 5)$  in the plane XOY. (b) Image of  $(-4, -3, 6)$  in the plane YOZ.

**Solution:**

The XOY plane is  $z = 0$ , so reflecting in it only flips the sign of the  $z$ -coordinate, since that plane acts as a mirror perpendicular to the  $z$ -axis alone;  $x, y$  are unaffected because they already lie in the mirror plane.

(a) Image =  $(-2, 3, -5)$ .

Similarly, the YOZ plane is  $x = 0$ , so reflection flips only the  $x$ -coordinate.

(b) Image =  $(4, -3, 6)$ .

**Final Answer**(a)  $(-2, 3, -5)$  (b)  $(4, -3, 6)$ 

**Common Mistake:** Flipping the wrong coordinate – remember XOY keeps  $x, y$  and flips  $z$ ; YOZ keeps  $y, z$  and flips  $x$ ; ZOX keeps  $z, x$  and flips  $y$ .

**Quick Recall:** Reflection in a coordinate plane just negates the one coordinate not named in that plane – no calculation needed beyond identifying which axis is "missing."

**Type 46**

Checking whether a given line is parallel to a given plane (i.e. the line's direction is perpendicular to the plane's normal), and finding an unknown constant for this

condition to hold, or for the line to actually lie in the plane.

**Formula Used:**  $\vec{b} \cdot \vec{n} = 0 \iff$  line parallel to (or lying in) the plane

Source: Exercise 11.5, Q5(ii)

**Given:** If the line  $\frac{x+1}{3} = \frac{y-2}{4} = \frac{z+6}{5}$  is parallel to the plane  $2x - 3y + kz = 0$ , find  $k$ .

**Solution:**

Parallel to the plane means the line's direction is perpendicular to the plane's normal, so their dot product must be zero – this is exactly the same perpendicularity condition used throughout the chapter, just applied between a line's DRs and a plane's normal DRs.

DRs of line  $\langle 3, 4, 5 \rangle$ ; normal DRs  $\langle 2, -3, k \rangle$ .

$$3(2) + 4(-3) + 5(k) = 0 \Rightarrow 6 - 12 + 5k = 0 \Rightarrow 5k = 6 \Rightarrow k = \frac{6}{5}.$$

**Final Answer**

$$k = \frac{6}{5}$$

*Note: If instead the question says the line lies in the plane (as in Q15(ii)), you need this same perpendicularity condition and a point on the line must satisfy the plane's equation – two conditions instead of one.*

**Common Mistake:** Applying the "lies in the plane" two-condition check when only "parallel to the plane" (one condition) was asked, or vice versa – read the question carefully, since parallel lines never touch the plane while a line lying in the plane does.

**Quick Recall:** Line parallel to a plane  $\iff$  dot product of line's DRs and plane's normal is 0; line *lying in* the plane needs this plus a point of the line satisfying the plane's equation.

### Type 47

Finding the point where a given line meets a given plane, by substituting the line's general (parametrised) point into the plane's equation.

Source: Exercise 11.5, Q16(ii)

**Given:** Find the coordinates of the point where the line through  $(3, -4, -5)$  and  $(2, -3, 1)$  crosses the plane  $2x + y + z = 7$ .

**Solution:**

DRs of the line =  $\langle -1, 1, 6 \rangle$ ; general point  $(3 - \lambda, -4 + \lambda, -5 + 6\lambda)$ .

Substituting into the plane's equation, since the crossing point must satisfy it:  $2(3 - \lambda) + (-4 + \lambda) + (-5 + 6\lambda) = 7 \Rightarrow 6 - 2\lambda - 4 + \lambda - 5 + 6\lambda = 7 \Rightarrow 5\lambda - 3 = 7 \Rightarrow \lambda = 2$ .

Substituting back:  $(3 - 2, -4 + 2, -5 + 12) = (1, -2, 7)$ .

### Final Answer

$(1, -2, 7)$

**Common Mistake:** Using the wrong of the two given points as the "base point" when writing the general point – either works as long as the DRs and substitution are done consistently, but mixing coordinates from both points is a common slip.

**Quick Recall:** Line meets a plane: write the line's general point with one parameter, plug into the plane's equation, solve for the parameter, substitute back.

### Type 48

Finding the intercepts a given plane makes on the coordinate axes, or – conversely – writing a plane's equation from its given intercepts, using the intercept form.

**Formula Used:**  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Source: Exercise 11.6, Q2(i)

**Given:** Find the intercepts made by the plane  $2x + y - z = 5$  on the coordinate axes.

**Solution:**

Divide throughout by 5 (the constant on the RHS) to force the equation into intercept form, since that form requires the RHS to be exactly 1:  $\frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1 \Rightarrow \frac{x}{5/2} + \frac{y}{5} + \frac{z}{-5} = 1$ .

Comparing with  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ , the intercepts are read directly off the denominators.

### Final Answer

Intercepts:  $a = \frac{5}{2}$ ,  $b = 5$ ,  $c = -5$

**Common Mistake:** Forgetting to divide the constant term into each term individually before comparing to intercept form – all three terms must have the same RHS of exactly 1.

**Quick Recall:** To find intercepts, divide the whole equation by the constant so the RHS becomes 1, then each denominator is the intercept on that axis; going the other way, intercepts  $a, b, c$  given  $\Rightarrow$  equation is  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$  directly.

**Type 49**

Finding the equation of a plane passing through the line of intersection of two given planes, and satisfying one extra condition (e.g. passing through the origin, through another point, or being parallel to something).

**Formula Used:**  $P_1 + \lambda P_2 = 0$

Source: Exercise 11.6, Q3(iii)

**Given:** Find the equation of the plane through the intersection of  $x + 2y + 3z + 4 = 0$  and  $x - y + z + 3 = 0$ , and passing through the origin.

**Solution:**

Any plane through the intersection of the two given planes can be written as a linear combination of them, since every point on the actual line of intersection satisfies both individually (hence any combination too), and this family sweeps out every plane containing that line as  $\lambda$  varies.

$$(x + 2y + 3z + 4) + \lambda(x - y + z + 3) = 0.$$

Since this plane must pass through the origin  $(0, 0, 0)$ , substitute and solve for  $\lambda$ :  $(0 + 0 + 0 + 4) + \lambda(0 - 0 + 0 + 3) = 0 \Rightarrow 4 + 3\lambda = 0 \Rightarrow \lambda = -\frac{4}{3}$ .

Substituting back and clearing fractions (multiply through by 3):  $3(x + 2y + 3z + 4) - 4(x - y + z + 3) = 0 \Rightarrow 3x + 6y + 9z + 12 - 4x + 4y - 4z - 12 = 0 \Rightarrow -x + 10y + 5z = 0$ , i.e.  $x - 10y - 5z = 0$ .

**Final Answer**

$$x - 10y - 5z = 0$$

**Common Mistake:** Forgetting to clear the fractional  $\lambda$  before finalising the equation – always multiply through to get whole-number coefficients as the last tidy-up step.

**Quick Recall:** Plane through intersection of two planes =  $P_1 + \lambda P_2 = 0$ ; use the one extra condition given to pin down  $\lambda$ .

**Type 50**

Finding the equation of a plane passing through a given point and parallel to two given lines – the plane's normal is the cross product of the two lines' direction vectors.

Source: Exercise 11.6, Q8

**Given:** Plane through  $(1, 2, -4)$  parallel to the lines  $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$  and  $\frac{x-1}{1} = \frac{y+3}{1} = \frac{z}{-1}$ .

**Solution:**

Since the plane is parallel to both lines, its normal must be perpendicular to both of their directions – exactly the cross-product idea used before, now applied to build a plane's normal instead of a new line's direction.

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 1 & 1 & -1 \end{vmatrix} = \hat{i}(3(-1) - 6(1)) - \hat{j}(2(-1) - 6(1)) + \hat{k}(2(1) - 3(1)) = \hat{i}(-9) - \hat{j}(-8) + \hat{k}(-1) = -9\hat{i} + 8\hat{j} - \hat{k}.$$

Using point-normal form through  $(1, 2, -4)$ :  $-9(x - 1) + 8(y - 2) - 1(z + 4) = 0 \Rightarrow -9x + 9 + 8y - 16 - z - 4 = 0 \Rightarrow -9x + 8y - z - 11 = 0$ , i.e.  $9x - 8y + z + 11 = 0$ .

**Final Answer**

$$9x - 8y + z + 11 = 0$$

**Common Mistake:** Using the points that the two given lines pass through (instead of only their direction vectors) inside the cross product – only the DRs matter for finding the normal.

**Quick Recall:** Plane parallel to two lines: normal = cross product of the two lines' direction vectors, then apply point-normal form.

**Type 51**

Finding the equation of a plane passing through a given point and containing (i.e. fully including) a given line.

Source: Exercise 11.6, Q5

**Given:** Plane through  $(3, 2, 0)$  containing the line  $\frac{x-3}{1} = \frac{y-6}{5} = \frac{z-4}{4}$ .

**Solution:**

The given line already passes through the point  $(3, 6, 4)$  with DRs  $\langle 1, 5, 4 \rangle$ . For the plane to *contain* this line, its normal must be perpendicular to both the line's direction *and* the vector joining the two known points on the plane (the given external point and the point on the line), so we cross these two vectors to get the normal.

Vector from  $(3, 2, 0)$  to  $(3, 6, 4)$ :  $\langle 0, 4, 4 \rangle$ .

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 5 & 4 \\ 0 & 4 & 4 \end{vmatrix} = \hat{i}(5(4) - 4(4)) - \hat{j}(1(4) - 4(0)) + \hat{k}(1(4) - 5(0)) = \hat{i}(4) - \hat{j}(4) + \hat{k}(4) = 4\hat{i} - 4\hat{j} + 4\hat{k}, \text{ i.e. } \langle 1, -1, 1 \rangle.$$

Using point-normal form through  $(3, 2, 0)$ :  $1(x - 3) - 1(y - 2) + 1(z - 0) = 0 \Rightarrow x - 3 - y + 2 + z = 0 \Rightarrow x - y + z - 1 = 0$ .

**Final Answer**

$$x - y + z - 1 = 0$$

**Common Mistake:** Using only the line's direction vector as the normal (as in Type 40) – that's wrong here, since the plane must also pass through a specific external point not on the line, requiring the second cross-product vector.

**Quick Recall:** Plane through a point, containing a line: cross the line's direction with the vector from the external point to any point on the line, to get the plane's normal.

### Type 52

Finding the equation of a plane that contains two given lines (whether parallel or intersecting).

Source: Exercise 11.6, Q11

**Given:** Vector equation of the plane containing the two parallel lines  $\frac{x-4}{1} = \frac{y-3}{-4} = \frac{z-2}{5}$  and  $\frac{x-3}{1} = \frac{y+2}{-4} = \frac{z}{5}$ .

#### Solution:

Since both lines share the same direction  $\langle 1, -4, 5 \rangle$  (they are parallel, not intersecting), we need a second vector lying in the plane: the vector joining one point from each line, e.g. from  $(4, 3, 2)$  to  $(3, -2, 0)$ :  $\langle -1, -5, -2 \rangle$ .

The plane's normal is the cross product of the shared direction and this connecting vector, since both vectors lie in the plane:

$$\begin{aligned}\vec{n} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -4 & 5 \\ -1 & -5 & -2 \end{vmatrix} = \hat{i}((-4)(-2) - 5(-5)) - \hat{j}(1(-2) - 5(-1)) + \hat{k}(1(-5) - (-4)(-1)) \\ &= \hat{i}(8 + 25) - \hat{j}(-2 + 5) + \hat{k}(-5 - 4) = 33\hat{i} - 3\hat{j} - 9\hat{k}, \text{ i.e. } \langle 11, -1, -3 \rangle \text{ after dividing by 3.} \\ \text{Using point-normal form through } (4, 3, 2): & 11(x - 4) - 1(y - 3) - 3(z - 2) = 0 \Rightarrow 11x - 44 - y + 3 - 3z + 6 = 0 \Rightarrow 11x - y - 3z - 35 = 0.\end{aligned}$$

#### Final Answer

$$11x - y - 3z - 35 = 0$$

**Common Mistake:** Trying to use only one direction vector (since both lines share it) as the plane's normal – one direction vector alone can never fully determine a plane's normal; a second, non-parallel vector (the connecting segment) is essential.

**Quick Recall:** Plane containing two parallel lines: cross the shared direction with the vector connecting one point from each line; for two intersecting lines, cross their two (different) direction vectors instead.

**Type 53**

Showing that two given lines are coplanar (using the scalar triple product test), and then finding the equation of the plane containing both.

**Formula Used:**  $[\vec{a}_2 - \vec{a}_1 \quad \vec{b}_1 \quad \vec{b}_2] = 0 \iff$  lines are coplanar

Source: Exercise 11.6, Q13

**Given:** Show the lines  $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$  and  $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$  are coplanar; find the equation of the plane containing them.

**Solution:**

Points:  $(-3, 1, 5)$  and  $(-1, 2, 5)$ ; directions  $\langle -3, 1, 5 \rangle$  and  $\langle -1, 2, 5 \rangle$ . Connecting vector  $= \langle -1 - (-3), 2 - 1, 5 - 5 \rangle = \langle 2, 1, 0 \rangle$ .

Coplanarity test – the scalar triple product of the connecting vector and both direction vectors must vanish:

$$\begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = 2(1(5) - 5(2)) - 1((-3)(5) - 5(-1)) + 0 = 2(5 - 10) - 1(-15 + 5) = 2(-5) - 1(-10) = -10 + 10 = 0.$$

Since this determinant is exactly zero, the lines are coplanar.

The plane's normal is then the cross product of the two direction vectors:  $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} =$

$$\hat{i}(5 - 10) - \hat{j}(-15 + 5) + \hat{k}(-6 + 1) = -5\hat{i} + 10\hat{j} - 5\hat{k}, \text{ i.e. } \langle 1, -2, 1 \rangle.$$

Using point-normal form through  $(-3, 1, 5)$ :  $1(x + 3) - 2(y - 1) + 1(z - 5) = 0 \Rightarrow x + 3 - 2y + 2 + z - 5 = 0 \Rightarrow x - 2y + z = 0$ .

**Final Answer**

Lines are coplanar; plane:  $x - 2y + z = 0$

**Common Mistake:** Confusing the coplanarity test (scalar triple product of connecting vector with *both* directions) with the plane's normal (cross product of *just* the two directions) – these are two different  $3 \times 3$  computations using overlapping but not identical vectors.

**Quick Recall:** Coplanar test: scalar triple product of (connecting vector, direction 1, direction 2) = 0; if coplanar, the plane's normal is the cross product of the two direction vectors alone.

**Type 54**

Finding the equation of a plane through three given points (either as coordinates or as position vectors), using the determinant form.

★ **Board Favorite** – Extremely common – "plane through 3 points" is one of the most frequently and directly examined skills in this whole topic.

**Formula Used:** 
$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

Source: Exercise 11.6, Q16(i)

**Given:** Find the equation of the plane through  $(-2, 6, 6)$ ,  $(1, -1, 0)$ ,  $(1, 2, -1)$ .

**Solution:**

Let  $A(-2, 6, 6)$ ,  $B(1, -1, 0)$ ,  $C(1, 2, -1)$ . We build two vectors in the plane –  $AB$  and  $AC$  – and use the determinant formula, since the plane's normal is exactly the cross product these vectors represent.

$$AB = \langle 3, -7, -6 \rangle, AC = \langle 3, -4, -7 \rangle.$$

$$\begin{vmatrix} x + 2 & y - 6 & z - 6 \\ 3 & -7 & -6 \\ 3 & -4 & -7 \end{vmatrix} = 0$$

Expanding along the top row:  $(x + 2)[(-7)(-7) - (-6)(-4)] - (y - 6)[3(-7) - (-6)(3)] + (z - 6)[3(-4) - (-7)(3)] = 0$

$$(x + 2)(49 - 24) - (y - 6)(-21 + 18) + (z - 6)(-12 + 21) = 0 \Rightarrow 25(x + 2) + 3(y - 6) + 9(z - 6) = 0$$

$$25x + 50 + 3y - 18 + 9z - 54 = 0 \Rightarrow 25x + 3y + 9z - 22 = 0.$$

**Final Answer**

$$25x + 3y + 9z - 22 = 0$$

**Common Mistake:** Sign errors while expanding the  $3 \times 3$  determinant – particularly missing the negative sign in front of the middle ( $y$ ) cofactor.

### Alternate Board-Style Example

Source: Exercise 11.6, Q17

**Given:** Plane through points with position vectors  $2\hat{i} + 5\hat{j} - 3\hat{k}$ ,  $-2\hat{i} - 3\hat{j} + 5\hat{k}$ ,  $5\hat{i} + 3\hat{j} - 3\hat{k}$ .

**Solution:**

Let these be A, B, C.  $AB = -4\hat{i} - 8\hat{j} + 8\hat{k}$ ,  $AC = 3\hat{i} - 2\hat{j} + 0\hat{k}$ .

$$\text{Normal} = AB \times AC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = \hat{i}(0 + 16) - \hat{j}(0 - 24) + \hat{k}(8 + 24) = 16\hat{i} + 24\hat{j} + 32\hat{k},$$

i.e.  $\langle 2, 3, 4 \rangle$  after dividing by 8.

Plane through A:  $2(x - 2) + 3(y - 5) + 4(z + 3) = 0 \Rightarrow 2x + 3y + 4z - 7 = 0$ , i.e.  $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$ .

**Final Answer**

$$\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = 7$$

**Quick Recall:** Plane through 3 points: form two vectors from one common point to the other two, cross them for the normal, then apply point-normal form (or use the full  $3 \times 3$  determinant shortcut directly).

**Type 55**

Checking whether four given points are coplanar (using the determinant/scalar-triple-product test), or finding an unknown value that makes four given points coplanar.

Source: Exercise 11.6, Q18(i)

**Given:** Show that  $(1, -1, 1)$ ,  $(2, 3, 1)$ ,  $(1, 2, 3)$ ,  $(0, -2, 3)$  are coplanar; find the equation of the plane.

**Solution:**

Take  $A(1, -1, 1)$  as the base point;  $AB = \langle 1, 4, 0 \rangle$ ,  $AC = \langle 0, 3, 2 \rangle$ ,  $AD = \langle -1, -1, 2 \rangle$ .

For all four points to be coplanar, the scalar triple product of  $AB$ ,  $AC$ ,  $AD$  must vanish, since this measures the (zero) volume of the parallelepiped they would otherwise form.

$$\begin{vmatrix} 1 & 4 & 0 \\ 0 & 3 & 2 \\ -1 & -1 & 2 \end{vmatrix} = 1(6 + 2) - 4(0 + 2) + 0 = 8 - 8 = 0.$$

Since the determinant is zero, the four points are coplanar. The plane itself is found using any three of the four points (say  $A$ ,  $B$ ,  $C$ ) with the standard three-point determinant

$$\text{method (Type 54): normal} = AB \times AC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 0 \\ 0 & 3 & 2 \end{vmatrix} = \hat{i}(8 - 0) - \hat{j}(2 - 0) + \hat{k}(3 - 0) = 8\hat{i} - 2\hat{j} + 3\hat{k}.$$

Plane through  $A$ :  $8(x - 1) - 2(y + 1) + 3(z - 1) = 0 \Rightarrow 8x - 2y + 3z - 13 = 0$ .

**Final Answer**

Points are coplanar; plane:  $8x - 2y + 3z - 13 = 0$

**Common Mistake:** Testing coplanarity using the wrong base point (any of the four works, but  $AB$ ,  $AC$ ,  $AD$  must all originate from the *same* chosen base point) – mixing base points invalidates the determinant.

**Quick Recall:** Four points coplanar  $\iff$  scalar triple product of any three vectors from a common base point among them is zero; once confirmed, build the actual plane from any three of the four points.

**Type 56**

Finding the equation of a plane with equal (or given-ratio) intercepts on the axes, or where the centroid of the triangle cut off by the plane on the axes is specified.

Source: Exercise 11.6, Q20(i)

**Given:** Plane through  $(2, 4, 6)$  making equal intercepts on all three axes.

**Solution:**

Equal intercepts means  $a = b = c$  in the intercept form  $\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$ , which simplifies to  $x + y + z = a$  – a single unknown constant to find using the given point.

Substituting  $(2, 4, 6)$ :  $2 + 4 + 6 = a \Rightarrow a = 12$ .

**Final Answer**

$$x + y + z = 12$$

*Note: For the centroid sub-case (Q21): if a plane meets the axes at  $A(a, 0, 0)$ ,  $B(0, b, 0)$ ,  $C(0, 0, c)$ , the centroid of  $\triangle ABC$  is  $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$  by the standard centroid formula – equate this to the given centroid to find  $a, b, c$ , then use the intercept form.*

**Common Mistake:** Forgetting that "equal intercepts" collapses all three unknowns  $a, b, c$  into just one unknown  $a$  – students sometimes still try to solve for three separate variables unnecessarily.

**Quick Recall:** Equal intercepts: plane becomes  $x + y + z = a$ , one unknown; given centroid of the intercept-triangle: intercepts are exactly 3 times the centroid's coordinates.

## V. Angle Between Planes & Distance Formulas

### Type 57

Finding the angle between two planes (vector or cartesian form), checking whether two planes are perpendicular, or finding an unknown constant for perpendicularity.

**Formula Used:**  $\cos \theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Source: Exercise 11.7, Q1(iv)

**Given:** Planes  $2x - y + z = 6$  and  $x + y + 2z = 7$ .

**Solution:**

Since the angle between two planes equals the angle between their normals, we read off the normal DRs directly:  $\langle 2, -1, 1 \rangle$  and  $\langle 1, 1, 2 \rangle$ .

$$\cos \theta = \frac{2(1) + (-1)(1) + 1(2)}{\sqrt{4 + 1 + 1}\sqrt{1 + 1 + 4}} = \frac{2 - 1 + 2}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2}.$$

**Final Answer**

$$\theta = 60^\circ$$

*Note: To instead find an unknown  $p$  making two planes perpendicular (as in Q3), simply set the dot product of the two normals to zero and solve – identical mechanics to line-perpendicularity, just applied to normal vectors.*

**Common Mistake:** Using the line-and-plane sine formula here by mistake – angle between two planes always uses  $\cos \theta$  on the two normals directly, never  $\sin \theta$ .

**Quick Recall:** Angle between planes = angle between their normals; read DRs of each normal straight off the coefficients and apply the cosine formula exactly as with two lines.

### Type 58

Finding the equation of a plane through a given point, parallel to a given plane – the new plane keeps the same normal, only the constant term changes.

★ **Board Favorite** – A short, elegant, and very frequently tested plane-construction shortcut – shows up constantly since it needs almost no computation once the pattern is recognised.

**Formula Used:**  $ax + by + cz = \lambda$

Source: Exercise 11.7, Q5(i)

**Given:** Plane through the point  $3\hat{i} + 4\hat{j} - \hat{k}$ , parallel to the plane  $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) + 5 = 0$ .

**Solution:**

A plane parallel to a given plane always keeps the exact same normal, and only the constant term is free to adjust for the new point – this is the whole idea of this type.

The general parallel plane is  $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = \lambda$  for some constant  $\lambda$  to be found.

Substituting the given point's position vector, since it must satisfy this new plane:  $(3\hat{i} + 4\hat{j} - \hat{k}) \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = \lambda \Rightarrow 6 - 12 - 5 = \lambda \Rightarrow \lambda = -11$ .

**Final Answer**

$$\vec{r} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) = -11$$

**Common Mistake:** Accidentally changing the normal's direction or magnitude while writing the new plane – the normal must be copied exactly from the given plane, only  $\lambda$  is unknown.

**Quick Recall:** Parallel plane through a new point: keep the same normal, substitute the new point to solve for the new constant term.

### Type 59

Finding the angle between a line and a plane (uses  $\sin \theta$ , not  $\cos \theta$ , since the formula measures the angle from the plane itself, not from the normal); also the reverse problem where the angle is given and an unknown constant must be found.

**Formula Used:**  $\sin \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$

Source: Exercise 11.7, Q7(i)

**Given:** Angle between the line  $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{2}$  and the plane  $3x + 4y + z + 5 = 0$ .

**Solution:**

Line DRs  $\langle 3, -1, 2 \rangle$ ; plane normal  $\langle 3, 4, 1 \rangle$ . Since the angle between a line and a plane is

measured from the plane surface (not from the normal), the formula uses  $\sin \theta$  instead of  $\cos \theta$  – this is the single biggest thing to remember about this type.

$$\sin \theta = \frac{|3(3) + (-1)(4) + 2(1)|}{\sqrt{9+1+4}\sqrt{9+16+1}} = \frac{|9-4+2|}{\sqrt{14}\sqrt{26}} = \frac{7}{\sqrt{364}} = \frac{7}{2\sqrt{91}}.$$

### Final Answer

$$\theta = \sin^{-1} \left( \frac{7}{2\sqrt{91}} \right)$$

*Note: Reverse sub-case (Q8): if the angle itself is given as  $\sin^{-1}(\lambda)$  or similar, substitute the known angle into the same formula and solve algebraically for the unknown constant appearing in the line's or plane's equation.*

**Common Mistake:** The single most common error in this entire chapter: using  $\cos \theta$  (the plane-to-plane or line-to-line formula) instead of  $\sin \theta$  for a line-and-plane angle.

**Quick Recall:** Line-plane angle uses  $\sin \theta$  (not  $\cos \theta$ ) on the dot product of the line's DRs and the plane's normal – this is the one formula in the whole chapter that breaks the "cosine" pattern.

### Type 60

Finding the equation of a plane through a given point, perpendicular to each of two other given planes – the required normal is the cross product of the two given planes' normals.

★ **Board Favorite** – A frequent ISC long-answer favourite – it neatly reuses the cross-product idea in yet another context (this time between two *normals* rather than two lines).

*Source: Exercise 11.7, Q13*

**Given:** Find the equation of the plane through  $(1, 1, -1)$  and perpendicular to the planes  $x + 2y + 3z = 7$  and  $2x - 3y + 4z = 0$ .

### Solution:

Since the required plane is perpendicular to both given planes, its own normal must be perpendicular to both of *their* normals – so the required normal is the cross product of  $\langle 1, 2, 3 \rangle$  and  $\langle 2, -3, 4 \rangle$ .

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix} = \hat{i}(8+9) - \hat{j}(4-6) + \hat{k}(-3-4) = 17\hat{i} + 2\hat{j} - 7\hat{k}.$$

Using point-normal form through  $(1, 1, -1)$ :  $17(x-1) + 2(y-1) - 7(z+1) = 0 \Rightarrow 17x - 17 + 2y - 2 - 7z - 7 = 0 \Rightarrow 17x + 2y - 7z - 26 = 0.$

**Final Answer**

$$17x + 2y - 7z - 26 = 0$$

**Common Mistake:** Using the given planes' equations' constant terms (the 7 and the 0) inside the cross product by mistake – the cross product uses only the normal DRs (coefficients of  $x, y, z$ ), never the constants.

**Quick Recall:** Plane perpendicular to two other planes: normal = cross product of the two given planes' normal vectors, then apply point-normal form.

**Type 61**

Finding the equation of a plane through two given points, perpendicular to another given plane; or through two points and parallel to a coordinate axis.

Source: Exercise 11.7, Q11

**Given:** Plane through  $(1, -1, 2)$ ,  $(2, -2, 2)$  and perpendicular to the plane  $6x - 2y + 2z = 9$ .

**Solution:**

Vector between the two points:  $\langle 1, -1, 0 \rangle$ . Since the required plane must also be perpendicular to the given plane, its normal is perpendicular to *that* plane's normal  $\langle 6, -2, 2 \rangle$  (i.e.  $\langle 3, -1, 1 \rangle$  simplified) as well as to the connecting vector – so we cross these two vectors together.

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 3 & -1 & 1 \end{vmatrix} = \hat{i}(-1 - 0) - \hat{j}(1 - 0) + \hat{k}(-1 + 3) = -\hat{i} - \hat{j} + 2\hat{k}.$$

Plane through  $(1, -1, 2)$ :  $-1(x - 1) - 1(y + 1) + 2(z - 2) = 0 \Rightarrow -x + 1 - y - 1 + 2z - 4 = 0 \Rightarrow -x - y + 2z - 4 = 0$ , i.e.  $x + y - 2z + 4 = 0$ .

**Final Answer**

$$x + y - 2z + 4 = 0$$

*Note: For the "parallel to a coordinate axis" sub-case, the axis direction (e.g.  $\langle 1, 0, 0 \rangle$  for the  $x$ -axis) simply plays the role of the second vector to cross with the connecting vector between the two points.*

**Common Mistake:** Crossing the connecting vector with the point coordinates of the given plane instead of with its normal DRs – always extract the normal from the given plane's equation first.

**Quick Recall:** Plane through two points, perpendicular to another plane: cross the vector joining the two points with the given plane's normal (or with a coordinate axis's direction, for the parallel-to-axis sub-case).

**Type 62**

Finding the perpendicular distance of a given point from a given plane (vector or cartesian form).

★ **Board Favorite** – The single most repeated numeric-answer question type across both plane sections – a guaranteed short-answer question almost every year.

**Formula Used:**  $d = \left| \frac{ax_1 + by_1 + cz_1 - k}{\sqrt{a^2 + b^2 + c^2}} \right|$

Source: Exercise 11.8, Q1(iii)

**Given:** Find the distance of  $(3, -2, 1)$  from the plane  $2x - y + 2z + 3 = 0$ .

**Solution:**

Substitute the point's coordinates directly into the plane's LHS expression, then divide by the magnitude of the normal, since this ratio is exactly the perpendicular distance formula.

$$d = \frac{|2(3) - (-2) + 2(1) + 3|}{\sqrt{4 + 1 + 4}} = \frac{|6 + 2 + 2 + 3|}{\sqrt{9}} = \frac{13}{3} \text{ units.}$$

**Final Answer**

$$d = \frac{13}{3} \text{ units}$$

**Common Mistake:** Forgetting the absolute value bars – the point could sit on either side of the plane, giving a negative numerator, but distance is always reported as positive.

**Alternate Board-Style Example**

Source: Exercise 11.8, Q2(ii)

**Given:** Length of perpendicular from origin to the plane  $\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) + 39 = 0$ .

**Solution:**

Rewriting in the standard  $\vec{r} \cdot \vec{n} = d'$  style:  $\vec{r} \cdot (3\hat{i} - 4\hat{j} - 12\hat{k}) = -39$ . For the origin,  $\vec{r} = \vec{0}$ , so we substitute directly and divide by  $|\vec{n}|$ .

$$d = \frac{|0 - (-39)|}{\sqrt{9 + 16 + 144}} = \frac{39}{\sqrt{169}} = \frac{39}{13} = 3 \text{ units.}$$

**Final Answer**

$$d = 3 \text{ units}$$

**Quick Recall:** Distance of a point from a plane: substitute the point into (normal)·(point) – constant, take absolute value, divide by the magnitude of the normal.

### Type 63

Finding the distance between two parallel planes, or the equation of the plane exactly midway between two given parallel planes.

**Formula Used:**  $d = \left| \frac{k_1 - k_2}{\sqrt{a^2 + b^2 + c^2}} \right|$

Source: Exercise 11.8, Q5(i)

**Given:** Find the distance between the planes  $2x + 3y + 4z = 4$  and  $4x + 6y + 8z = 12$ .

**Solution:**

The second plane's equation must first be scaled so its normal matches the first plane's exactly, since the distance formula only applies to planes with the same (or proportional, then normalised) normal.

Dividing the second equation by 2:  $2x + 3y + 4z = 6$ .

Now both planes share normal  $\langle 2, 3, 4 \rangle$  with constants  $k_1 = 4$  and  $k_2 = 6$ .

$$d = \frac{|4 - 6|}{\sqrt{4 + 9 + 16}} = \frac{2}{\sqrt{29}} \text{ units.}$$

### Final Answer

$$d = \frac{2}{\sqrt{29}} \text{ units}$$

*Note: For the mid-parallel plane sub-case, average the two constants (after scaling to the same normal): the mid-plane is  $ax + by + cz = \frac{k_1 + k_2}{2}$ .*

**Common Mistake:** Applying the distance formula directly to the two planes' original (unscaled) constants without first checking the normals are identical, not just proportional – always scale one equation to match the other's normal exactly.

**Quick Recall:** Distance between parallel planes: scale both equations to share the exact same normal, then it's just  $\frac{|k_1 - k_2|}{|\vec{n}|}$ ; mid-parallel plane simply averages  $k_1$  and  $k_2$ .

### Type 64

Finding the distance of a point from a plane, but measured not perpendicularly – instead along a given line's direction (i.e. following a specified line from the point

until it meets the plane).

Source: Exercise 11.8, Q20(ii)

**Given:** Find the distance of the point  $(2, 3, 4)$  measured along the line  $\frac{x-4}{3} = \frac{y+5}{6} = \frac{z+1}{2}$  from the plane  $3x + 2y + 2z + 5 = 0$ .

**Solution:**

Unlike the perpendicular-distance type, here we must travel specifically along the given line's direction (not the plane's normal) until we reach the plane – so we parametrise using the line's DCs (to make the parameter a genuine distance) starting from the given point.

DRs of the line  $\langle 3, 6, 2 \rangle$ ; magnitude  $= \sqrt{9 + 36 + 4} = \sqrt{49} = 7$ , so DCs are  $\langle \frac{3}{7}, \frac{6}{7}, \frac{2}{7} \rangle$ .

A general point at distance  $r$  from  $(2, 3, 4)$  along this direction is  $(2 + \frac{3}{7}r, 3 + \frac{6}{7}r, 4 + \frac{2}{7}r)$ .

Substituting into the plane's equation, since this point must eventually land exactly on the plane:  $3(2 + \frac{3}{7}r) + 2(3 + \frac{6}{7}r) + 2(4 + \frac{2}{7}r) + 5 = 0$

$$6 + \frac{9}{7}r + 6 + \frac{12}{7}r + 8 + \frac{4}{7}r + 5 = 0 \Rightarrow 25 + \frac{25}{7}r = 0 \Rightarrow r = -7.$$

Taking the magnitude, the required distance is  $|r| = 7$  units.

**Final Answer**

7 units

**Common Mistake:** Using the plane's normal direction (as in ordinary perpendicular distance) instead of the given line's own direction – this type specifically requires travelling along the stated line, which is not perpendicular to the plane in general.

**Quick Recall:** Distance "measured along a line": parametrise using that line's DCs (not the plane's normal) from the point, substitute into the plane's equation, solve for  $r$ , and take its magnitude.